



US 20250283480A1

(19) **United States**(12) **Patent Application Publication****Kray et al.**(10) **Pub. No.: US 2025/0283480 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **TURBINE ENGINE WITH COMPOSITE AIRFOILS**

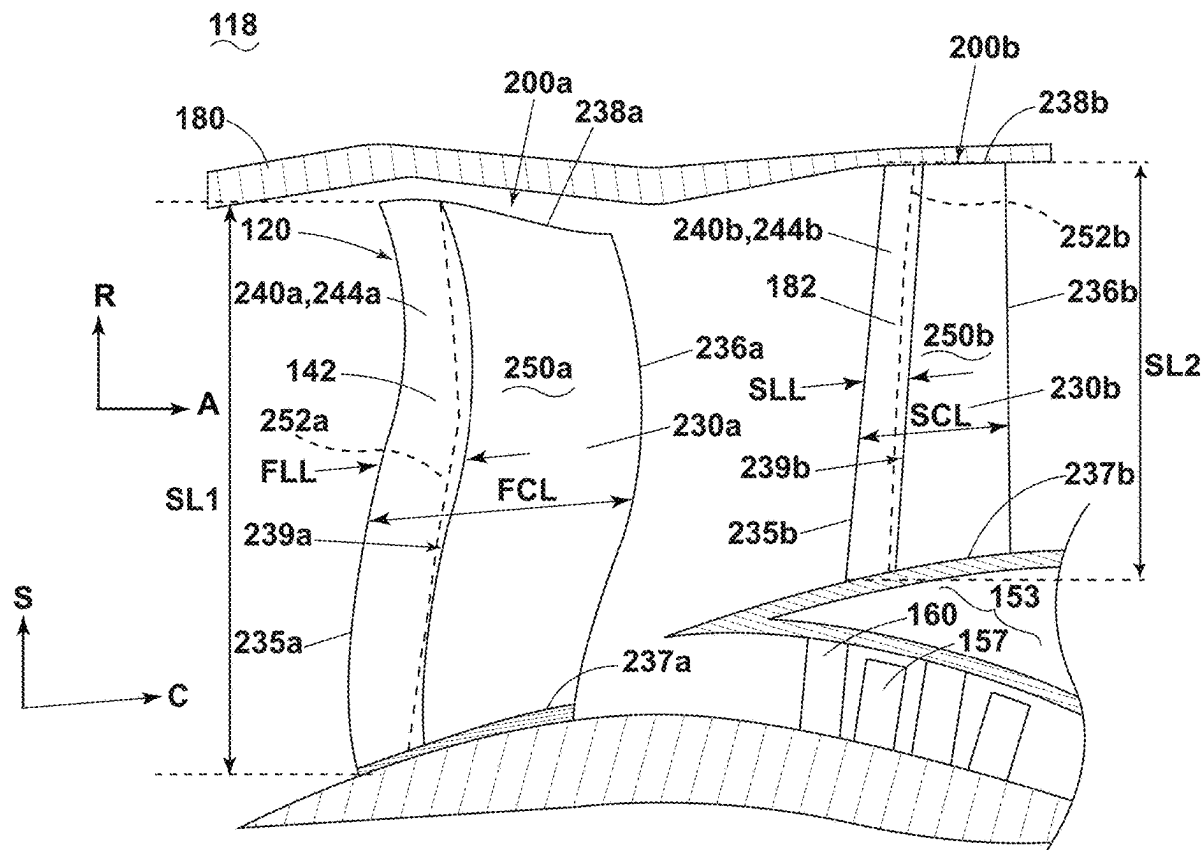
which is a continuation-in-part of application No. 18/171,533, filed on Feb. 20, 2023, now Pat. No. 12,158,082.

(71) Applicant: **GENERAL ELECTRIC COMPANY,**
Evendale, OH (US)(72) Inventors: **Nicholas Joseph Kray,** Mason, OH (US); **Arthur William Sibbach,** Boxford, MA (US)(21) Appl. No.: **19/214,354**(22) Filed: **May 21, 2025****Related U.S. Application Data**

(63) Continuation of application No. 18/792,952, filed on Aug. 2, 2024, now Pat. No. 12,352,284, which is a continuation-in-part of application No. 18/741,419, filed on Jun. 12, 2024, now Pat. No. 12,326,154,

Publication Classification(51) **Int. Cl.**
F04D 29/32 (2006.01)(52) **U.S. Cl.**
CPC **F04D 29/324** (2013.01); **F05D 2240/303** (2013.01); **F05D 2240/304** (2013.01); **F05D 2300/603** (2013.01)(57) **ABSTRACT**

A turbine engine with an engine core defining an engine centerline and comprising a rotor and a stator. The turbine engine including a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor. An airfoil in the set of composite airfoils including a composite portion extending chordwise between a composite leading edge and a trailing edge and a leading edge protector coupled to the composite portion.



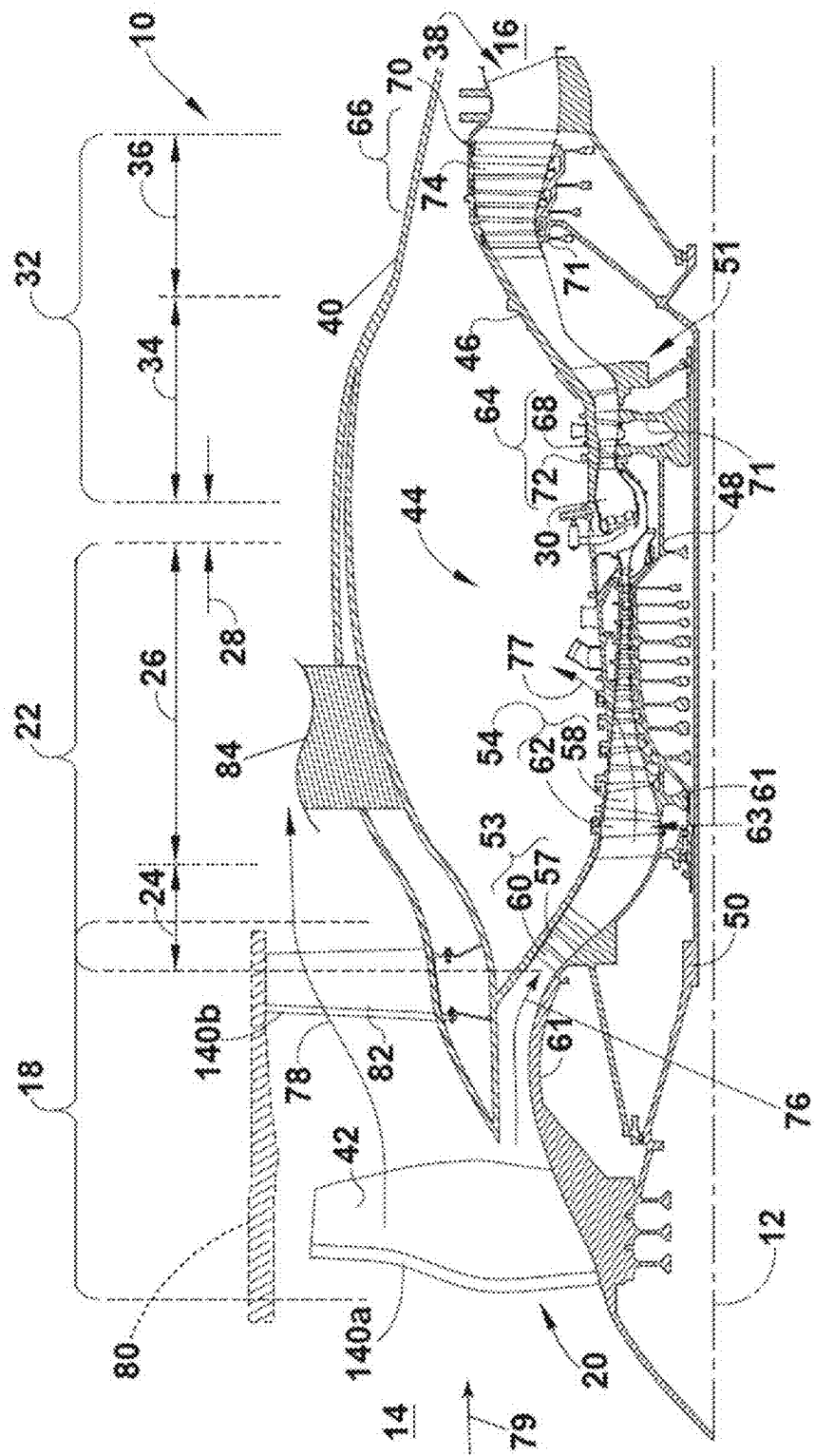


FIG. 1

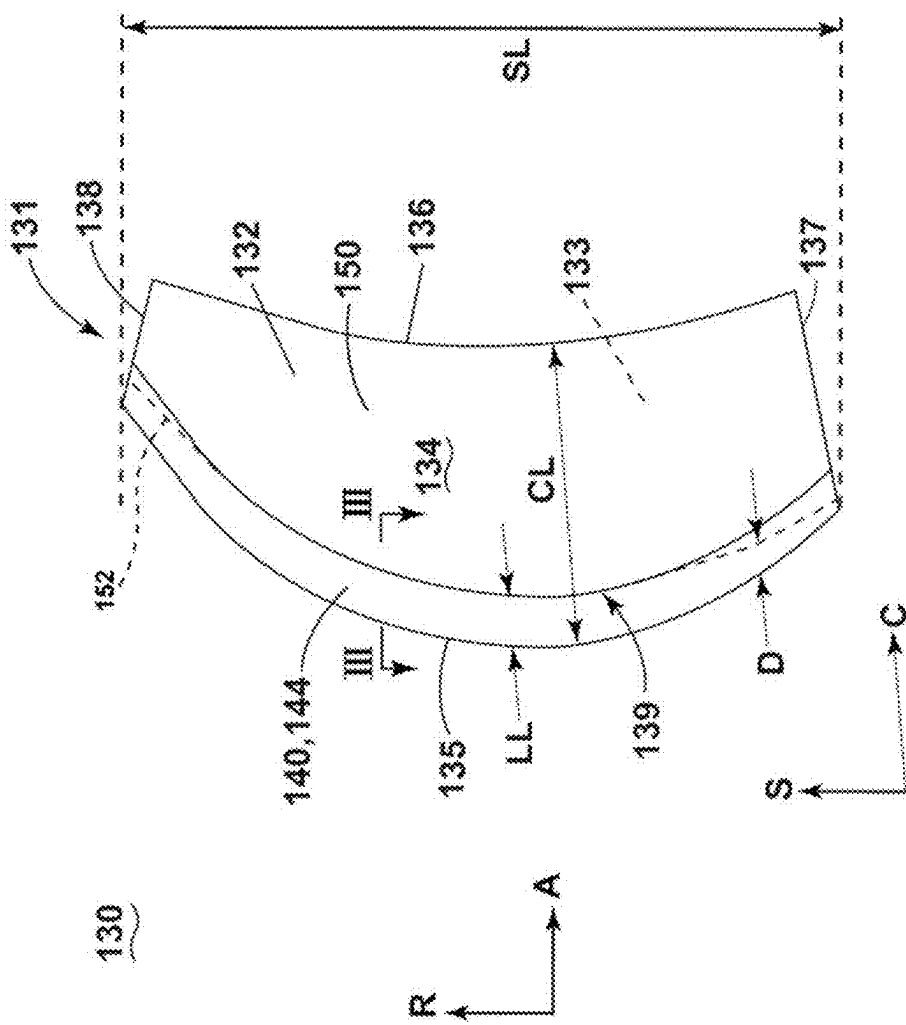


FIG. 2

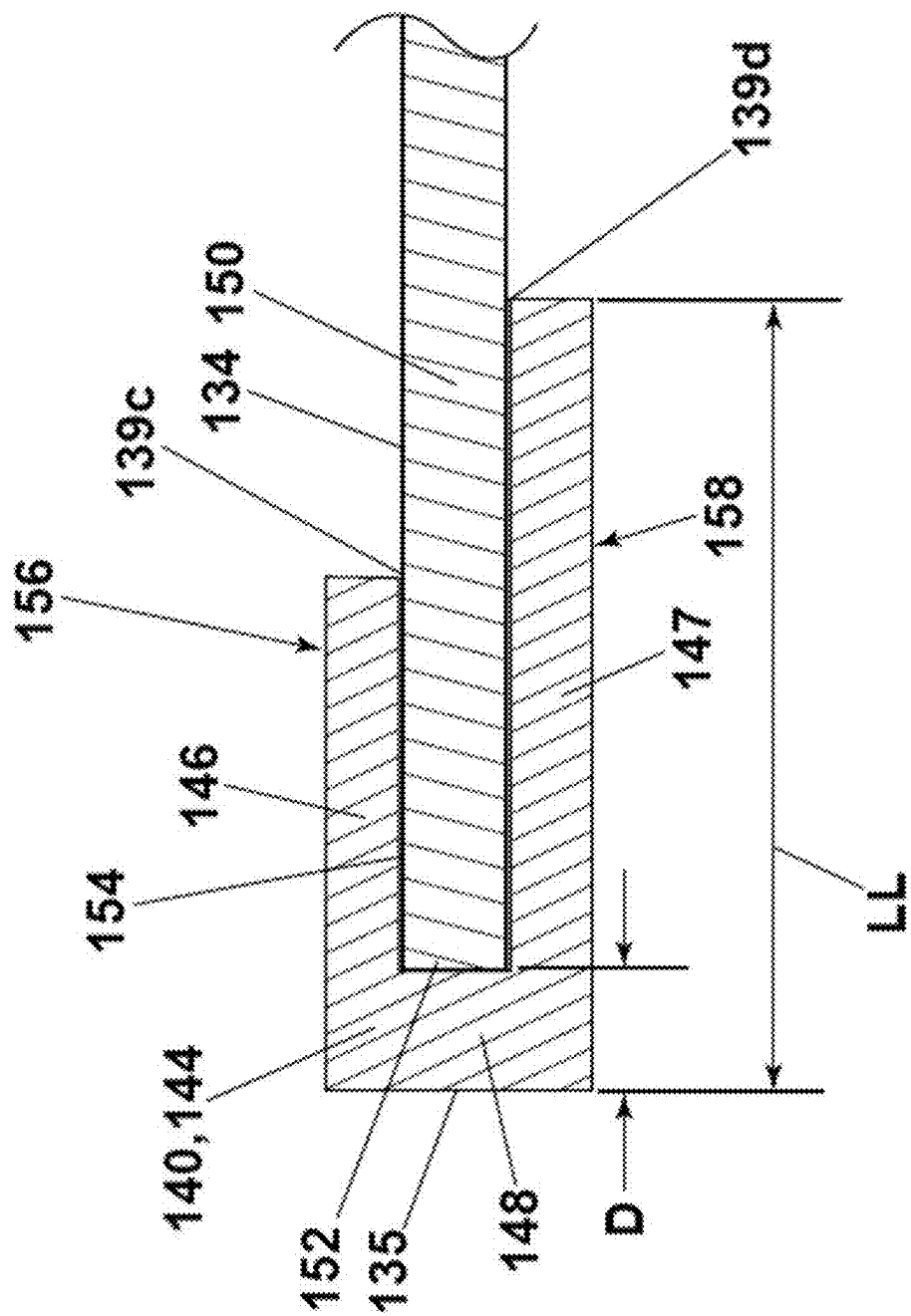
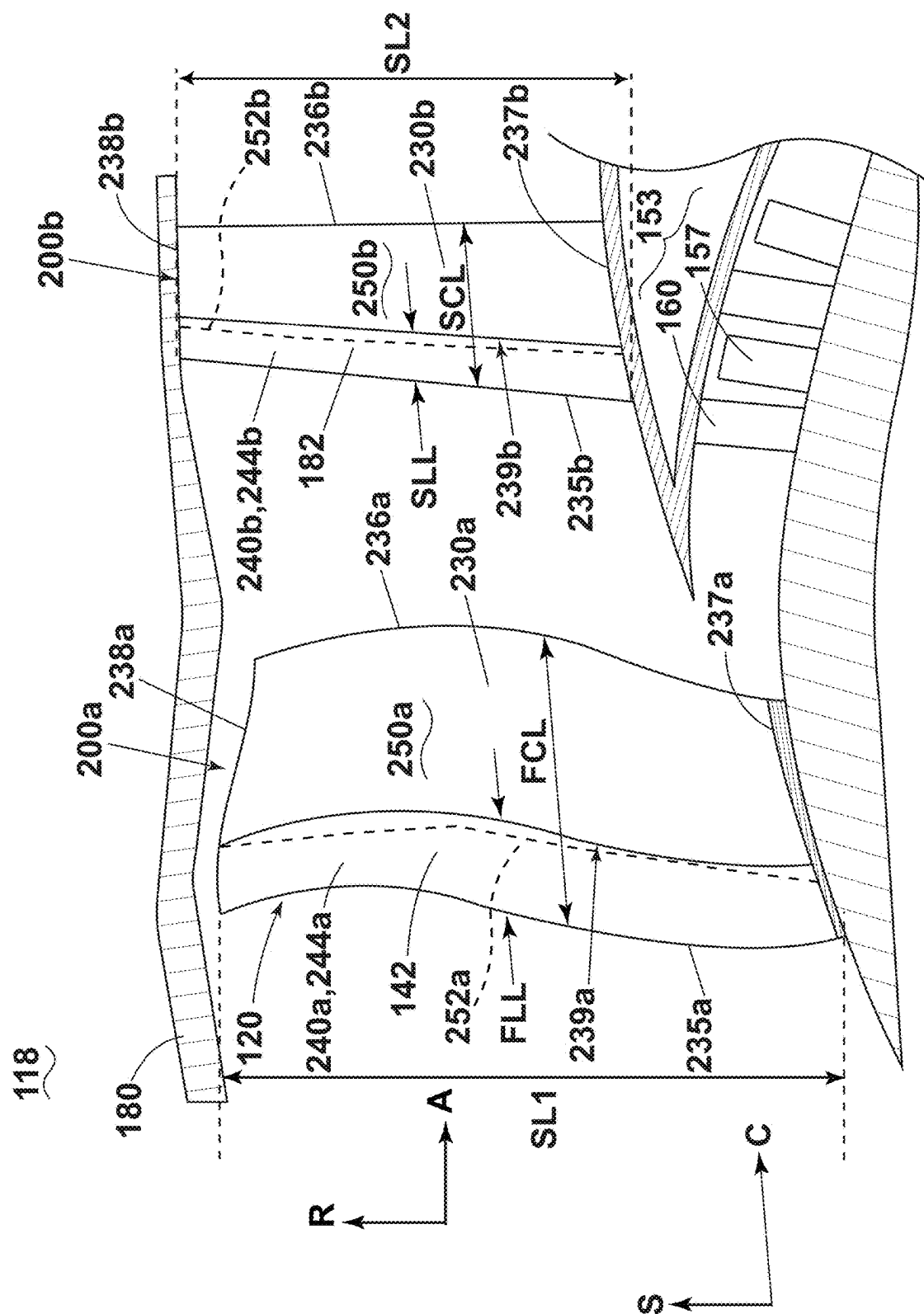


FIG. 3



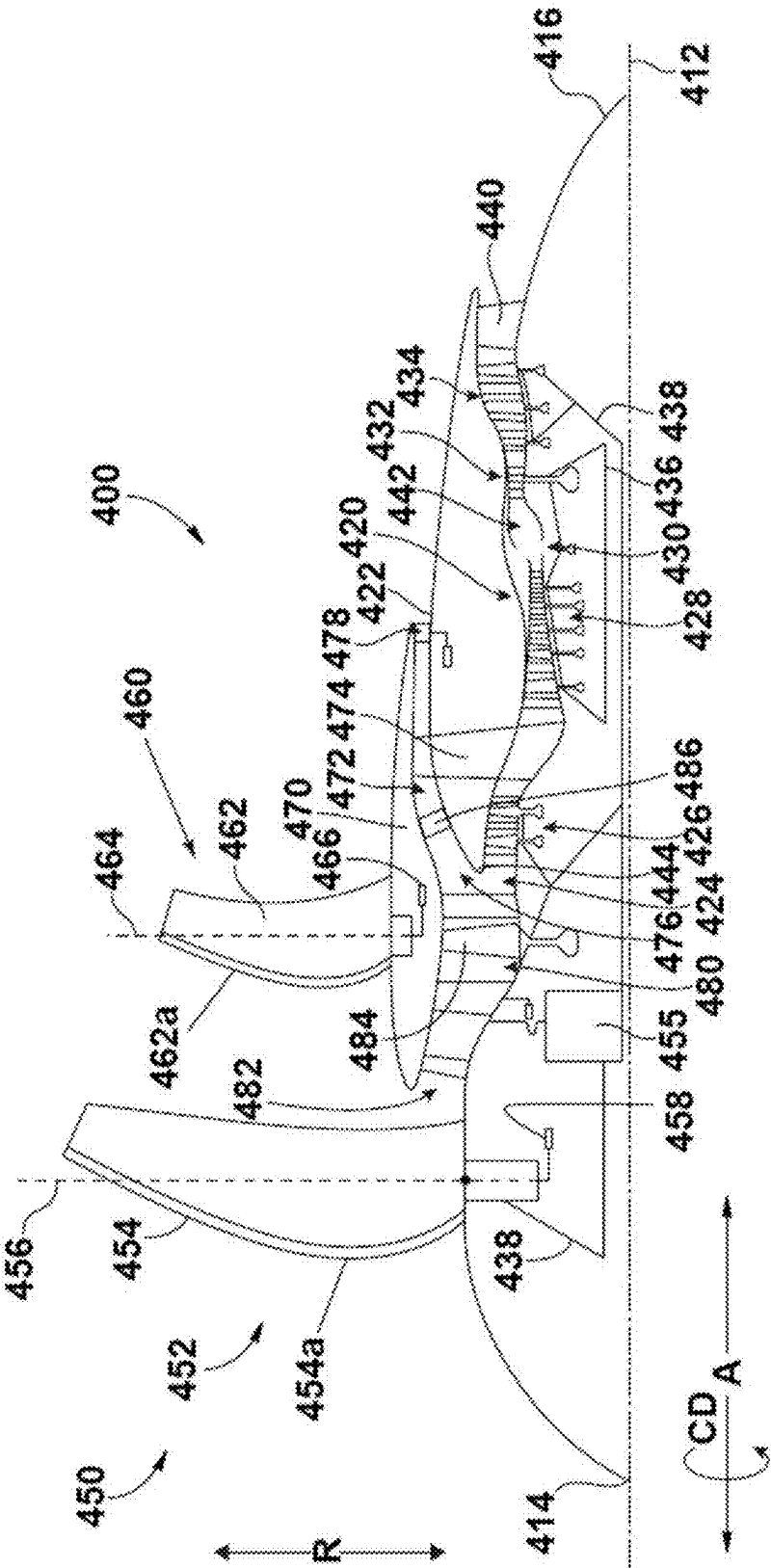


FIG. 6

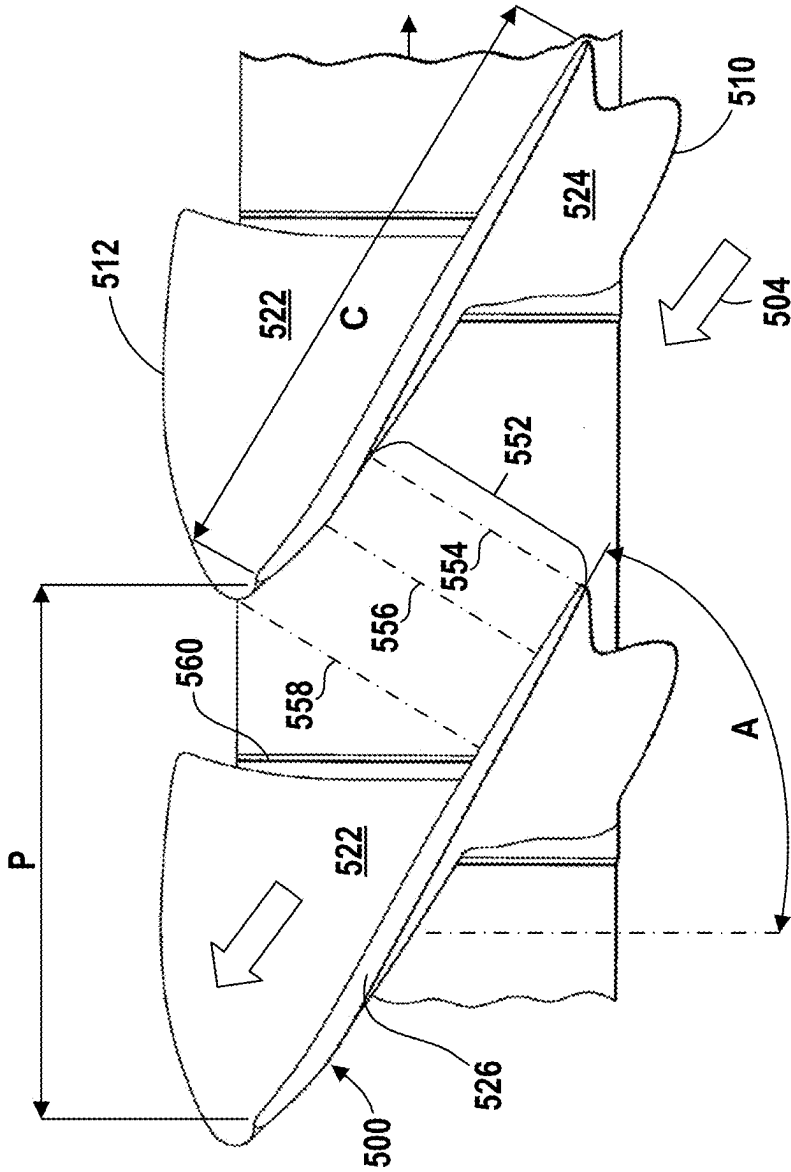


FIG. 7

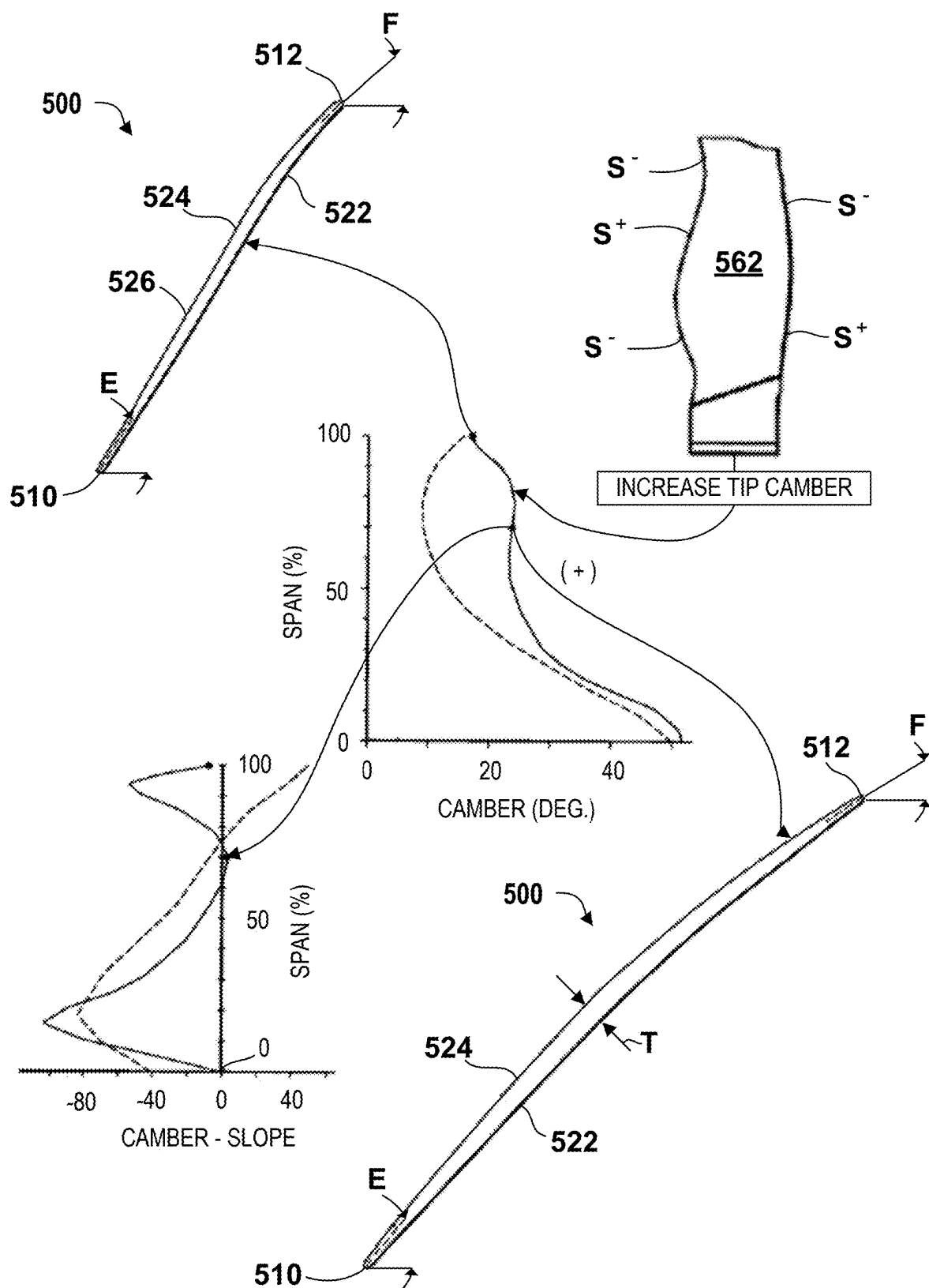


FIG. 8

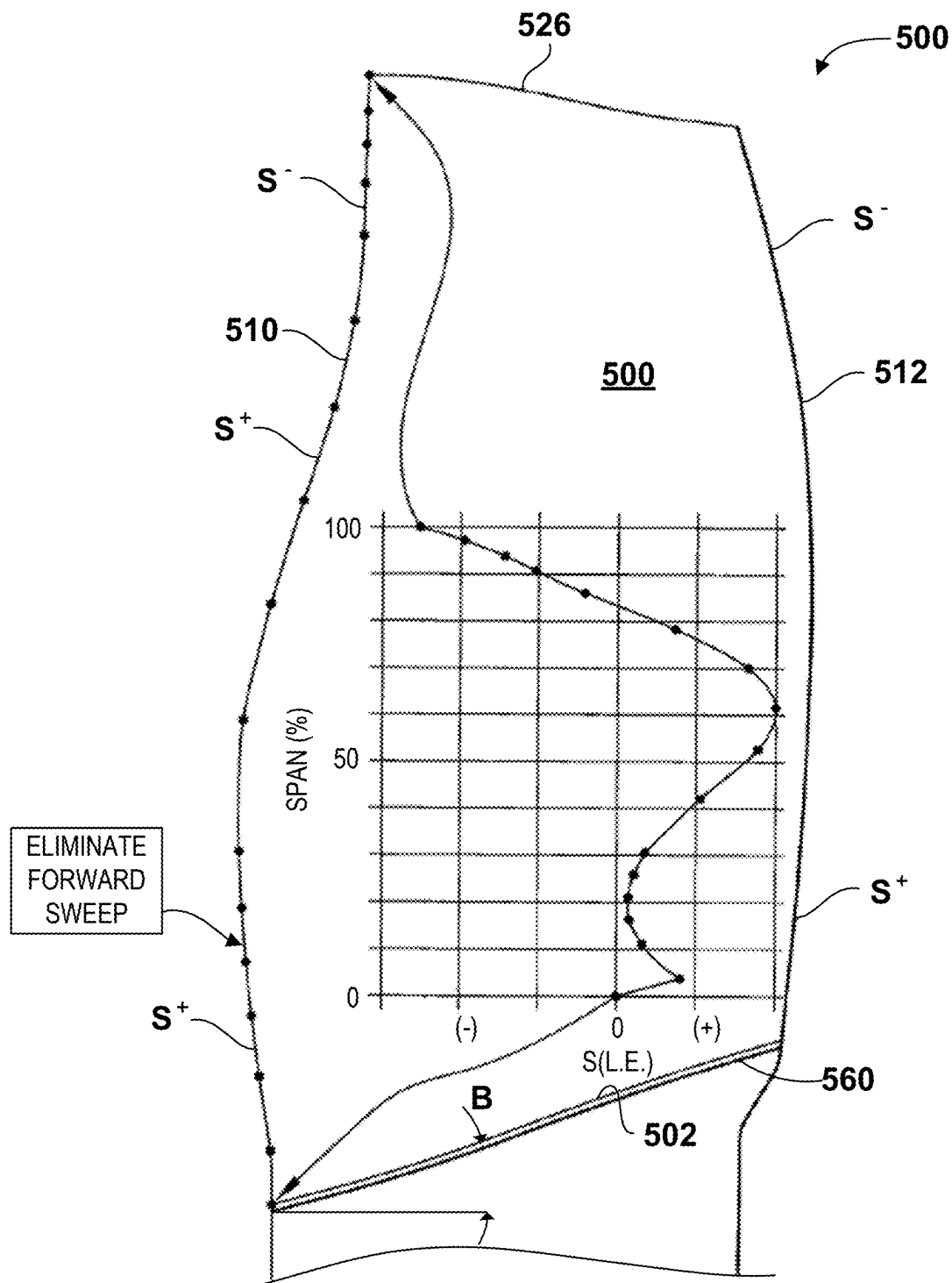


FIG. 9

TURBINE ENGINE WITH COMPOSITE AIRFOILS

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application is a continuation of U.S. patent application Ser. No. 18/792,952, filed Aug. 2, 2024, now allowed, which is a continuation-in-part of U.S. application Ser. No. 18/741,419, filed Jun. 12, 2024, now allowed, which is a continuation-in-part of U.S. application Ser. No. 18/171,533, filed Feb. 20, 2023, now U.S. Pat. No. 12,158,082, which issued on Dec. 3, 2024, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The disclosure generally relates to a component for a turbine engine, more specifically, to a composite airfoil.

BACKGROUND

[0003] Composite materials typically include a fiber-reinforced matrix and exhibit a high strength to weight ratio. Due to the high strength to weight ratio and moldability to adopt relatively complex shapes, composite materials are utilized in various applications, such as a turbine engine or an aircraft. Composite materials can be, for example, installed on or define a portion of the fuselage and/or wings, rudder, manifold, airfoil, or other components of the aircraft or turbine engine. Extreme loading or sudden forces can be applied to the composite components of the aircraft or turbine engine. For example, extreme loading can occur to one or more airfoils during ingestion of various materials by the turbine engine.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0005] FIG. 1 is a schematic cross-sectional view of a turbine engine in accordance with an exemplary embodiment of the present disclosure.

[0006] FIG. 2 is a schematic illustration of a composite airfoil in the form of a fan blade for the turbine engine of FIG. 1 according to an exemplary embodiment of the present disclosure.

[0007] FIG. 3 is a schematic cross-section taken along line III-III of FIG. 2.

[0008] FIG. 4 is a schematic enlarged view of an exemplary fan section for the turbine engine of FIG. 1 according to an exemplary embodiment of the present disclosure.

[0009] FIG. 5 is a schematic cross-sectional view of a gas turbine engine in accordance with another exemplary embodiment of the present disclosure.

[0010] FIG. 6 is a schematic cross-sectional view of a gas turbine engine in accordance with another exemplary embodiment of the present disclosure.

[0011] FIG. 7 is a top plan view of two adjacent airfoils having staggers that increase from the roots to the tips of the airfoils.

[0012] FIG. 8 illustrates two scaled graphs of airfoil camber and slope thereof versus span.

[0013] FIG. 9 is a side elevational view of an airfoil and a graph of corresponding aerodynamic sweep thereof.

DETAILED DESCRIPTION

[0014] Aspects of the disclosure herein are directed to a plurality of composite airfoil stages. For purposes of illustration, the present disclosure will be described with respect to the plurality of composite airfoil stages within an engine being a first stage of airfoils in the form of fan blades and a second stage of airfoils immediately downstream the first stage of airfoils as an outlet guide vane (OGV). While fan blades and OGVs are illustrated, it should be understood that any consecutive sets of stages are contemplated. Further, it will be understood, that aspects of the disclosure herein are not so limited and may have general applicability within an engine, including compressors, as well as in non-aircraft applications, such as other mobile applications and non-mobile industrial, commercial, and residential applications.

[0015] Reference will now be made in detail to composite fan blades and composite OGVs, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings.

[0016] The term “composite,” as used herein is, is indicative of a material that does not include metal material. A composite can be a combination of at least two or more non-metallic elements or materials. Examples of a composite material can be, but not limited to, a polymer matrix composite (PMC), a ceramic matrix composite (CMC), carbon fibers, a polymeric resin, a thermoplastic, bismaleimide (BMI), a polyimide materials, an epoxy resin, glass fibers, and silicon matrix materials.

[0017] As used herein, a “composite” component refers to a structure or a component including any suitable composite material. Composite components, such as a composite airfoil, can include several layers or plies of composite material. The layers or plies can vary in stiffness, material, and dimension to achieve the desired composite component or composite portion of a component having a predetermined weight, size, stiffness, and strength.

[0018] One or more layers of adhesive can be used in forming or coupling composite components. Adhesives can include resin and phenolics, wherein the adhesive can require curing at elevated temperatures or other hardening techniques.

[0019] As used herein, PMC refers to a class of materials. By way of example, the PMC material is defined in part by a prepreg, which is a reinforcement material pre-impregnated with a polymer matrix material, such as thermoplastic resin. Non-limiting examples of processes for producing thermoplastic prepregs include hot melt pre-pregging in which the fiber reinforcement material is drawn through a molten bath of resin and powder pre-pregging in which a resin is deposited onto the fiber reinforcement material, by way of non-limiting example electrostatically, and then adhered to the fiber, by way of non-limiting example, in an oven or with the assistance of heated rollers. The prepregs can be in the form of unidirectional tapes or woven fabrics, which are then stacked on top of one another to create the number of stacked plies desired for the part.

[0020] Multiple layers of prepreg are stacked to the proper thickness and orientation for the composite component and then the resin is cured and solidified to render a fiber reinforced composite part. Resins for matrix materials of PMCs can be generally classified as thermosets or thermoplastics. Thermoplastic resins are generally categorized as polymers that can be repeatedly softened and flowed when

heated and hardened when sufficiently cooled due to physical rather than chemical changes. Notable example classes of thermoplastic resins include nylons, thermoplastic polyesters, polyaryletherketones, and polycarbonate resins. Specific example of high performance thermoplastic resins that have been contemplated for use in aerospace applications include, polyetheretherketone (PEEK), polyetherketoneketone (PEKK), polyetherimide (PEI), polyaryletherketone (PAEK), and polyphenylene sulfide (PPS). In contrast, once fully cured into a hard rigid solid, thermoset resins do not undergo significant softening when heated, but instead thermally decompose when sufficiently heated. Notable examples of thermoset resins include epoxy, bismaleimide (BMI), and polyimide resins.

[0021] Instead of using a prepreg, in another non-limiting example, with the use of thermoplastic polymers, it is possible to utilize a woven fabric. Woven fabric can include, but is not limited to, dry carbon fibers woven together with thermoplastic polymer fibers or filaments. Non-prepreg braided architectures can be made in a similar fashion. With this approach, it is possible to tailor the fiber volume of the part by dictating the relative concentrations of the thermoplastic fibers and reinforcement fibers that have been woven or braided together. Additionally, different types of reinforcement fibers can be braided or woven together in various concentrations to tailor the properties of the part. For example, glass fibers, carbon fibers, and thermoplastic fibers could all be woven together in various concentrations to tailor the properties of the part. The carbon fibers provide the strength of the system, the glass fibers can be incorporated to enhance the impact properties, which is a design characteristic for parts located near the inlet of the engine, and the thermoplastic fibers provide the binding for the reinforcement fibers.

[0022] In yet another non-limiting example, resin transfer molding (RTM) can be used to form at least a portion of a composite component. Generally, RTM includes the application of dry fibers or matrix material to a mold or cavity. The dry fibers or matrix material can include prepreg, braided material, woven material, or any combination thereof.

[0023] Resin can be pumped into or otherwise provided to the mold or cavity to impregnate the dry fibers or matrix material. The combination of the impregnated fibers or matrix material and the resin are then cured and removed from the mold. When removed from the mold, the composite component can require post-curing processing.

[0024] It is contemplated that RTM can be a vacuum assisted process. That is, the air from the cavity or mold can be removed and replaced by the resin prior to heating or curing. It is further contemplated that the placement of the dry fibers or matrix material can be manual or automated.

[0025] The dry fibers or matrix material can be contoured to shape the composite component or direct the resin. Optionally, additional layers or reinforcing layers of material differing from the dry fiber or matrix material can also be included or added prior to heating or curing.

[0026] As used herein, CMC refers to a class of materials with reinforcing fibers in a ceramic matrix. Generally, the reinforcing fibers provide structural integrity to the ceramic matrix. Some examples of reinforcing fibers can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), non-oxide carbon-based materials (e.g., carbon), oxide ceramics

(e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al_2O_3), silicon dioxide (SiO_2), aluminosilicates such as mullite, or mixtures thereof), or mixtures thereof.

[0027] Some examples of ceramic matrix materials can include, but are not limited to, non-oxide silicon-based materials (e.g., silicon carbide, silicon nitride, or mixtures thereof), oxide ceramics (e.g., silicon oxycarbides, silicon oxynitrides, aluminum oxide (Al_2O_3), silicon dioxide (SiO_2), aluminosilicates, or mixtures thereof), or mixtures thereof. Optionally, ceramic particles (e.g., oxides of Si, Al, Zr, Y, and combinations thereof) and inorganic fillers (e.g., pyrophyllite, wollastonite, mica, talc, kyanite, and montmorillonite) can also be included within the ceramic matrix.

[0028] Generally, particular CMCs can be referred to as their combination of type of fiber/type of matrix. For example, C/SiC for carbon-fiber-reinforced silicon carbide, SiC/SiC for silicon carbide-fiber-reinforced silicon carbide, SiC/SiN for silicon carbide fiber-reinforced silicon nitride, SiC/SiC-SiN for silicon carbide fiber-reinforced silicon carbide/silicon nitride matrix mixture, etc. In other examples, the CMCs can be comprised of a matrix and reinforcing fibers comprising oxide-based materials such as aluminum oxide (Al_2O_3), silicon dioxide (SiO_2), aluminosilicates, and mixtures thereof. Aluminosilicates can include crystalline materials such as mullite ($3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$), as well as glassy aluminosilicates.

[0029] In certain non-limiting examples, the reinforcing fibers may be bundled and/or coated prior to inclusion within the ceramic matrix. For example, bundles of the fibers may be formed as a reinforced tape, such as a unidirectional reinforced tape. A plurality of the tapes may be laid up together to form a preform component. The bundles of fibers may be impregnated with a slurry composition prior to forming the preform or after formation of the preform. The preform may then undergo thermal processing, and subsequent chemical processing, to arrive at a component formed of a CMC material having a desired chemical composition. For example, the preform may undergo a cure or burn-out to yield a high char residue in the preform, and subsequent melt-infiltration with silicon, or a cure or pyrolysis to yield a silicon carbide matrix in the preform, and subsequent chemical vapor infiltration with silicon carbide. Additional steps may be taken to improve densification of the preform, either before or after chemical vapor infiltration, by injecting it with a liquid resin or polymer followed by a thermal processing step to fill the voids with silicon carbide. CMC material as used herein may be formed using any known or hereinafter developed methods including but not limited to melt infiltration, chemical vapor infiltration, polymer impregnation pyrolysis (PIP), or any combination thereof.

[0030] Such materials, along with certain monolithic ceramics (i.e., ceramic materials without a reinforcing material), are particularly suitable for higher temperature applications. Additionally, these ceramic materials are lightweight compared to superalloys, yet can still provide strength and durability to the component made therefrom. Therefore, such materials are currently being considered for many gas turbine components used in higher temperature sections of gas turbine engines, such as airfoils (e.g., turbines, and vanes), combustors, shrouds and other like components, that would benefit from the lighter-weight and higher temperature capability these materials can offer.

[0031] The term “metallic” as used herein is indicative of a material that includes metal such as, but not limited to, titanium, iron, aluminum, stainless steel, and nickel alloys. A metallic material or alloy can be a combination of at least two or more elements or materials, where at least one is a metal.

[0032] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0033] As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0034] As used herein, the term “upstream” refers to a direction that is opposite the fluid flow direction, and the term “downstream” refers to a direction that is in the same direction as the fluid flow. The term “fore” or “forward” means in front of something and “aft” or “rearward” means behind something. For example, with regard to a turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

[0035] The term “fluid” may be a gas or a liquid, or multi-phase. The term “fluid communication” means that a fluid is capable of making the connection between the areas specified.

[0036] Additionally, as used herein, the terms “radial” or “radially” refer to a direction away from a common center. For example, in the overall context of a turbine engine, radial refers to a direction along a ray extending between a center longitudinal axis of the engine and an outer engine circumference.

[0037] All directional references (e.g., radial, axial, proximal, distal, upper, lower, upward, downward, left, right, lateral, front, back, top, bottom, above, below, vertical, horizontal, clockwise, counterclockwise, upstream, downstream, forward, aft, etc.) are only used for identification purposes to aid the reader’s understanding of the present disclosure, and do not create limitations, particularly as to the position, orientation, or use of aspects of the disclosure described herein. Connection references (e.g., attached, coupled, connected, and joined) are to be construed broadly and can include intermediate structural elements between a collection of elements and relative movement between elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and in fixed relation to one another. The exemplary drawings are for purposes of illustration only and the dimensions, positions, order and relative sizes reflected in the drawings attached hereto can vary.

[0038] The singular forms “a”, “an”, and “the” include plural references unless the context clearly dictates otherwise. Furthermore, as used herein, the term “set” or a “set” of elements can be any number of elements, including only one.

[0039] Leading length or “LL” as used herein refers to a length between a leading edge of the airfoil and a seam between a leading edge protector and a portion of the airfoil.

[0040] First leading length or “FLL” as used herein refers to the leading length of a first stage of airfoils.

[0041] Second leading length or “SLL” as used herein refers to the leading length of a second stage of airfoils immediately downstream from the first stage of airfoils.

[0042] Chord length or “CL” as used herein refers to a length between a leading edge of the airfoil and a trailing edge of the airfoil.

[0043] First chord length or “FCL” as used herein refers to the chord length of the first stage of airfoils.

[0044] Second chord length or “SCL” as used herein refers to the chord length of the second stage of airfoils.

[0045] Airfoil protection factor or “APF” as used herein refers to a relationship in the form of a ratio of the leading length to the chord length of the airfoil. As more protection is provided for any given airfoil, the leading length increases and in turn so does the APF.

[0046] Stage performance factor or “SPF” as used herein refers to a relationship in the form of a ratio of the airfoil protection factor for the first stage of airfoils, or “APF1” to the airfoil protection factor for the second stage of airfoils, or “APF2”.

[0047] Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0048] In certain exemplary embodiments of the present disclosure, a turbine engine defining a centerline and a circumferential direction is provided. The turbine engine may generally include a turbomachine and a rotor assembly. The rotor assembly may be driven by the turbomachine. The turbomachine, the rotor assembly, or both may define a substantially annular flow path relative to the centerline of the turbine engine. In certain aspects of the present disclosure, an unducted or open rotor turbine engine includes a set of circumferentially spaced fan blades, which extend, exteriorly, beyond a nacelle encasing or engine core.

[0049] The turbine engine includes airfoils in the form of blades and vanes. The airfoils described herein can be a plurality of airfoils provided circumferentially about the centerline or be partially provided about a portion of the centerline. At least one airfoil in the plurality of airfoils includes a protective covering on a leading edge of the airfoil. The protective covering can be a metal covering. The protective covering is referred to herein as leading edge protector.

[0050] The leading edge protector can be designed for various flight conditions, including take off, descent, and idle. The objective, when designing an airfoil, specifically a composite fan blade and a composite outlet guide vane can be generally stated as balancing an added weight component from the protective covering, or sheath, on the leading edge with an acceptable amount of protection of the leading edge. The balancing of efficient weight designs can be particularly important in large turbofan applications of traditional direct drive, gear-reduction designs, and open-rotor designs. Key factors to consider include that the ratio of the leading edge chord to the blade chord is a balance between the leading edge dominating the response to a bird ingestion or similar event, and the PMC airfoil dominating the characteristics of the blade aerodynamics in normal operation.

[0051] There is a tradeoff between the percent of the airfoil chord that is covered by the leading edge protector, and the

performance of the airfoil. The protective covering provides a stiffness to the airfoil for bird ingestion, but the remainder of the blade is desirable to be flexible for aerodynamic purposes. Because the fan blade rotates and the OGV is stationary the dynamics of a bird ingestion event differs for the two airfoils.

[0052] The inventors have determined that the leading edge protector must overlap with enough of the composite airfoil in order to provide a strong enough bond, but it is desirable to minimize the overlap in order for the composite blade to flex. The leading edge protector also provides erosion protection to a composite airfoil and is required for both static and rotating airfoils. The leading edge protector characteristics have been developed from multiple tests and simulation analyses covering the ingestion of birds of varying sizes at varying span positions, and analysis of blades that have been returned for repair following bird strikes in revenue service. Furthermore, the OGV is not rotating, and experiences a different stress when impacted by a bird after it has passed through the fan. Multiple simulations and analyses depending on how the bird strikes the fan, whether it hits directly centered on a leading edge protector or hits between two adjacent blades where it is more likely to pass through without being sliced into smaller pieces. The relationship between the percent chord of a rotating and non-rotating blade that is covered by the leading edge protector is not obvious due to the difference in the forces acting upon the airfoils when struck by a bird or similar object.

[0053] The inventors' practice has proceeded in the manner of designing airfoil stages, modifying the airfoil stages with the addition of the leading edge protector, and redesigning the airfoil stages with the leading edge protector meeting protection requirements associated with the airfoil stages. After calculating and checking an amount of protection provided and an amount of weight increase or decreases associated with the leading edge protector, the process is repeated for other stages of during the design of several different types of turbomachines, such as those shown in FIG. 1. In other words, an airfoil design can meet performance requirements for one location in the engine, but not necessarily for another location.

[0054] FIG. 1 is a schematic cross-sectional diagram of a turbine engine 10 for an aircraft. By way of non-limiting example the turbine engine 10 is illustrated as an open rotor turbine engine. The turbine engine 10 has a generally longitudinally extending axis or engine centerline 12 extending from a forward end 14 to an aft end 16. The turbine engine 10 includes, in downstream serial flow relationship, a set of circumferentially spaced blades or propellers defining a fan section 18 including a fan 20, a compressor section 22 including a booster or low pressure (LP) compressor 24 and a high pressure (HP) compressor 26, a combustion section 28 including a combustor 30, a turbine section 32 including a HP turbine 34, and a LP turbine 36, and an exhaust section 38. The turbine engine 10 as described herein is meant as a non-limiting example, and other architectures are possible, such as, but not limited to, a steam turbine engine, a supercritical carbon dioxide turbine engine, or any other suitable turbine engine.

[0055] An exterior surface, defined by a nacelle 40, of the turbine engine 10 extends from the forward end 14 of the turbine engine 10 toward the aft end 16 of the turbine engine 10 and covers at least a portion of the compressor section 22, the combustion section 28, the turbine section 32, and the

exhaust section 38. The fan section 18 can be positioned at a forward portion of the nacelle 40 and extend radially outward from the nacelle 40 of the turbine engine 10, specifically, the fan section 18 extends radially outward from the nacelle 40. The fan section 18 includes a set of fan blades 42, and a set of outlet guide vanes (OGV) 82 downstream the set of fan blades 42, both disposed radially about the engine centerline 12. The turbine engine 10 includes any number of one or more sets of rotating blades or propellers (e.g., the set of fan blades 42) disposed upstream of a set of stationary fan vanes (e.g., the set of OGVs 82). As a non-limiting example, the turbine engine 10 can include multiple sets of rotating blades and stationary vanes. The set of fan blades 42 can include a first leading edge protector 140a and the set of OGVs 82 can include a second leading edge protector 140b. As such, the turbine engine 10 is further defined as an unducted single-fan turbine engine. The turbine engine 10 is further defined by the location of the fan section 18 with respect to the combustion section 28. The fan section 18 can be upstream, downstream, or in-line with the axial positioning of the combustion section 28. In some aspects of the disclosure herein, the turbine engine can include a fan casing 80 (shown in dotted line) surrounding the fan 20 to define a ducted turbine engine.

[0056] The compressor section 22, the combustion section 28, and the turbine section 32 are collectively referred to as an engine core 44, which generates combustion gases. The engine core 44 is surrounded by an engine casing 46, which is operatively coupled with a portion of the nacelle 40 of the turbine engine 10.

[0057] A HP shaft or spool 48 disposed coaxially about the engine centerline 12 of the turbine engine 10 drivingly connects the HP turbine 34 to the HP compressor 26. A LP shaft or spool 50, which is disposed coaxially about the engine centerline 12 of the turbine engine 10 within the larger diameter annular HP spool 48, drivingly connects the LP turbine 36 to the LP compressor 24 and fan 20. The HP spools 48 and the LP spool 50 are rotatable about the engine centerline 12 and couple to a set of rotatable elements, which collectively define a rotor 51.

[0058] It will be appreciated that the turbine engine 10 is either a direct drive or an integral drive engine utilizing a reduction gearbox coupling the LP shaft or spool 50 to the fan 20.

[0059] The LP compressor 24 and the HP compressor 26, respectively, include a set of compressor stages 53, 54, in which a set of compressor blades 57, 58 rotate relative to a corresponding set of static compressor vanes 60, 62 (also called a nozzle) to compress or pressurize the stream of fluid passing through the stage. In a single compressor stage 53, 54, multiple compressor blades 57, 58 are provided in a ring and extend radially outward relative to the engine centerline 12, from a blade platform to a blade tip, while the corresponding static compressor vanes 60, 62 are positioned upstream of and adjacent to the compressor blades 57, 58. It is noted that the number of blades, vanes, and compressor stages shown in FIG. 1 were selected for illustrative purposes only, and that other numbers are possible.

[0060] The compressor blades 57, 58 for a stage of the compressor are mounted to a disk 61, which is mounted to the corresponding one of the HP and LP spools 48, 50, with each stage having its own disk 61. The static compressor vanes 60, 62 for a stage of the compressor are mounted to the engine casing 46 in a circumferential arrangement.

[0061] The HP turbine 34 and the LP turbine 36, respectively, include a set of turbine stages 64, 66, in which a set of turbine blades 68, 70 are rotated relative to a corresponding set of static turbine vanes 72, 74 (also called a nozzle) to extract energy from the stream of fluid passing through the stage. In a single turbine stage 64, 66, multiple turbine blades 68, 70 are provided in a ring and extends radially outward relative to the engine centerline 12, from a blade platform to a blade tip, while the corresponding static turbine vanes 72, 74 are positioned upstream of and adjacent to the turbine blades 68, 70. It is noted that the number of blades, vanes, and turbine stages shown in FIG. 1 were selected for illustrative purposes only, and that other numbers are possible.

[0062] The turbine blades 68, 70 for a stage of the turbine are mounted to a disk 71, which is mounted to the corresponding one of the HP and LP spools 48, 50, with each stage having a dedicated disk 71. The static turbine vanes 72, 74 for a stage of the compressor are be mounted to the engine casing 46 in a circumferential arrangement. The compressor blades 57, 58 and the turbine blades 68, 70 described herein can be part of a blisk, rather than being mounted to a disk.

[0063] Complementary to the rotor portion, the stationary portions of the turbine engine 10, such as the static vanes 60, 62, 72, 74 among the compressor section 22 and the turbine section 32 are also referred to individually or collectively as a stator 63. As such, the stator 63 refers to the combination of non-rotating elements throughout the turbine engine 10.

[0064] The nacelle 40 is operatively coupled to the turbine engine 10 and covers at least a portion of the engine core 44, the engine casing 46, or the exhaust section 38. At least a portion of the nacelle 40 extends axially forward or upstream the illustrated portion. For example, the nacelle 40 extends axially forward such that a portion of the nacelle 40 overlays or covers a portion of the fan section 18 or a booster section (not illustrated) of the turbine engine 10. A pylon 84 mounts the turbine engine 10 to an exterior structure (e.g., a fuselage of an aircraft, a wing, a tail wing, etc.).

[0065] During operation of the turbine engine 10, a freestream airflow 79 flows against a forward portion of the turbine engine 10. A portion of the freestream airflow 79 becomes an inlet airflow 78. The inlet airflow 78 flows through the set of fan blades 42 and over the nacelle 40 of the turbine engine 10. Subsequently, the inlet airflow 78 flows over at least a portion of the set of OGVs 82, which directs the inlet airflow 78 such that it is transverse toward the engine centerline 12. The inlet airflow 78 then flows past the set of OGVs 82, following the curvature of the nacelle 40 and toward the exhaust section 38.

[0066] A portion of the freestream airflow 79 enters the engine core 44 after flowing through the set of fan blades 42 and is described as a working airflow 76, which is used for combustion within the engine core 44. More specifically, the working airflow 76 flows into the LP compressor 24, which then pressurizes the working airflow 76 thus defining a pressurized airflow that is supplied to the HP compressor 26, which further pressurizes the air. The working airflow 76, or the pressurized airflow, from the HP compressor 26 is mixed with fuel in the combustor 30 and ignited, thereby generating combustion gases. Some work is extracted from these gases by the HP turbine 34, which drives the HP compressor 26. The combustion gases are discharged into the LP turbine 36, which extracts additional work to drive the LP compres-

sor 24, and the working airflow 76, or exhaust gas, is ultimately discharged from the turbine engine 10 via the exhaust section 38. The driving of the LP turbine 36 drives the LP spool 50 to rotate the fan 20 and the LP compressor 24. The working airflow 76, including the pressurized airflow and the combustion gases, defines a working airflow that flows through the compressor section 22, the combustion section 28, and the turbine section 32 of the turbine engine 10.

[0067] The working airflow 76 and at least some of the inlet airflow 78 merge downstream of the exhaust section 38 of the turbine engine 10. The working airflow 76 and the inlet airflow 78, together, form an overall thrust of the turbine engine 10.

[0068] It is contemplated that a portion of the working airflow 76 is drawn as bleed 77 air (e.g., from the compressor section 22). The bleed air 77 provides an airflow to engine components requiring cooling. The temperature of the working airflow 76 exiting the combustor 30 is significantly increased with respect to the working airflow 76 within the compressor section 22. As such, cooling provided by the bleed air 77 is necessary for operating of such engine components in the heightened temperature environments or a hot portion of the turbine engine 10. In the context of a turbine engine, the hot portions of the engine are normally downstream of the combustor 30, especially the turbine section 32, with the HP turbine 34 being the hottest portion as it is directly downstream of the combustion section 28. Other sources of cooling fluid are, but are not limited to, fluid discharged from the LP compressor 24 or the HP compressor 26.

[0069] FIG. 2 is schematic illustration of a composite airfoil 130 in the form of, by way of non-limiting example, a fan blade 131. The fan blade 131 can be, by way of non-limiting example, a blade of the set of fan blades 42 or a blade from the compressor blades 57, 58 or the turbine blades 68, 70. Further, the composite airfoil 130 can be a vane of the set of OGVs 82 or a vane of the static vanes 60, 62, 72, 74. It is contemplated that the composite airfoil 130 can be a blade, vane, airfoil, or other component of any turbine engine, such as, but not limited to, a gas turbine engine, a turboprop engine, a turboshaft engine, or a turbofan engine.

[0070] The composite airfoil 130 can include a wall 132 bounding an interior 133. The wall 132 can define an exterior surface 134 extending radially between a leading edge 135 and a trailing edge 136 to define a chordwise direction (denoted "C"). The composite airfoil 130 has a chord length (denoted "CL") measured along the chordwise direction C between the leading edge 135 and the trailing edge 136. The exterior surface 134 can further extend between a root 137 and a tip 138 to define a spanwise direction (denoted "S"). The composite airfoil 130 has a span length (denoted "SL") measured along the spanwise direction S between the root 137 and the tip 138 where the root is considered 0% of the span length SL and the tip 138 is considered 100% of the span length SL. The span length SL is the maximum distance between the root 137 and the tip 138 of the composite airfoil 130. It will be understood that the composite airfoil 130 can take any suitable shape, profile, or form including that the leading edge 135 need not be curved.

[0071] An axial direction (denoted "A") extends generally across the page from right to left. The axial direction A is

parallel to the engine centerline 12 (FIG. 1). A radial direction (denoted “R”) extends perpendicularly away from the axial direction A. It should be understood that the spanwise direction S is parallel to the radial direction R. The chordwise direction C can extend generally along the axial direction A, however with more bend in the composite airfoil 130, it should be understood that the chordwise direction C can extend both into and out of the page and across the page from left to right.

[0072] The exterior surface 134 is defined by a leading edge protector 140 and a composite portion 150. A seam 139, separates the leading edge protector 140 from the composite portion 150 along the exterior surface 134. The leading edge protector 140 extends along the chordwise direction C between the leading edge 135 and the seam 139 to define a leading length (denoted “LL”).

[0073] The leading edge protector 140 is typically a metallic leading edge protector and can be made of, but is not limited to, steel, aluminum, refractory metals such as titanium, or superalloys based on nickel, cobalt, or iron. It should be understood that the leading edge protector 140 for the fan blade 131 can be a metallic leading edge protector while a set of stationary vanes downstream from the fan blade 131, by way of non-limiting example the set of OGVs 82 (FIG. 1), have the second leading edge protector 140b (FIG. 1) made of a polyurethane material. Further, the leading edge protectors 140, 140a, 140b described herein can be any suitable material such as metal, thermoplastic, or polyurethane, where both are the same, or different.

[0074] The composite portion 150 can include a composite leading edge 152 spaced a distance (denoted “D”) from the leading edge 135. The composite leading edge 152 can define at least a portion of, or all of the seam 139. It is further contemplated that at least a part of the leading edge protector 140 overlaps the composite portion 150 such that at least a portion of, illustrated in dashed line, or all of the composite leading edge 152 is located upstream from the seam 139. In other words, the leading edge protector 140 can define a sheath 144 on the composite leading edge 152.

[0075] The composite portion 150 can be made of one or more layers of material. The one or more layers of material can be applied during the same stage or different stages of the manufacturing of the composite airfoil 130. By way of non-limiting example, composite portion 150 can include at least a polymer matrix composite (PMC) portion or a polymeric portion. The polymer matrix composite can include, but is not limited to, a matrix of thermoset (epoxies, phenolics) or thermoplastic (polycarbonate, polyvinylchloride, nylon, acrylics) and embedded glass, carbon, steel, or Kevlar fibers.

[0076] The leading edge protector 140 and the composite portion 150 can be formed by a variety of methods, including additive manufacturing, casting, electroforming, or direct metal laser melting, in non-limiting examples. As used herein, an “additively manufactured” component refers to a component formed by an additive manufacturing (AM) process, wherein the component is built layer-by-layer by successive deposition of material. AM is an appropriate name to describe the technologies that build 3D objects by adding layer-upon-layer of material, whether the material is plastic, ceramic, or metal. AM technologies can utilize a computer, 3D modeling software (Computer Aided Design or CAD), machine equipment, and layering material. Once a CAD sketch is produced, the AM equipment can read in

data from the CAD file and lay down or add successive layers of liquid, powder, sheet material or other material, in a layer-upon-layer fashion to fabricate a 3D object. It should be understood that the term “additive manufacturing” encompasses many technologies including subsets like 3D Printing, Rapid Prototyping (RP), Direct Digital Manufacturing (DDM), layered manufacturing and additive fabrication. Non-limiting examples of additive manufacturing that can be utilized to form an additively-manufactured component include powder bed fusion, vat photopolymerization, binder jetting, material extrusion, directed energy deposition, material jetting, or sheet lamination. It is also contemplated that a process utilized could include printing a negative of the part, either by a refractory metal, ceramic, or printing a plastic, and then using that negative to cast the component.

[0077] It will be shown herein that a relationship between the leading length LL and the chord length CL can be referred to herein as an airfoil protection factor or simply as “APF”. In other words, for any given composite airfoil 130 having a predetermined chord length CL, an amount of coverage provided by the leading edge protector 140 increases, so does the leading length LL and in turn the APF.

[0078] FIG. 3 is a schematic cross-section taken along line III-III of FIG. 2. The leading edge protector 140 is the sheath 144 with a first wall 146, a second wall 147, and a third wall 148 interconnecting the first wall 146 and the second wall 147. The first wall 146, second wall 147, and third wall 148 of the leading edge protector 140 are oriented and shaped such that they define a generally U-shaped (or C-shaped) channel 154 therebetween. As shown in FIG. 3 and as will be discussed below, the channel 154 is sized and shaped to receive the composite leading edge 152 of the composite portion 150. Notably, the shape of the channel 154 is shown by way of example only and the channel 154 is not limited to this specific shape and is not drawn to scale.

[0079] The composite airfoil 130 can extend between a first side 156 and a second side 158. The seam 139 can be two seams 139c, 139d at corresponding ends of the channel 154. The leading length LL is measured from the leading edge 135 to the seam 139d furthest from the leading edge 135. While illustrated at two different locations, it should be understood that the seams 139c, 139d can be located at the same leading length LL. While illustrated as rectangular blunt ends at the seam 139, the leading edge protector 140 can taper such that the leading edge protector 140 and the composite portion 150 are flush to define the exterior surface 134.

[0080] FIG. 4 is schematic enlarged view of a fan section 118 similar to fan section 18 therefore, like parts of the fan section 118 (FIG. 1) will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the fan section 18 applies to the fan section 118, except where noted.

[0081] A set of compressor stages 153 include a set of compressor blades 157 rotating relative to a corresponding set of static compressor vanes 160. A set of fan blades 142 define a fan section 118 including a fan 120. The turbine engine can include a fan casing 180 surrounding the fan 120.

[0082] The set of fan blades 142 defines a first stage of airfoils 200a within the fan section 118 (FIG. 1). A first airfoil 230a in the first stage of airfoils 200a is similar to the previously described airfoil 130, therefore like parts of the first airfoil 230a will be identified with like numerals

increased by 100 and having a notation “a” with it being understood that the description of the like parts of the airfoil **130** applies to the first airfoil **230a**, except where noted. While only a single fan blade is shown in the cross-section it will be understood that the set of fan blades **142** are included and spaced about the fan section **118**.

[0083] The first airfoil **230a** has a first span length (denoted “SL1”) measured along the spanwise direction S between a first root **237a** and a first tip **238a** where the first root **237a** is considered 0% of the first span length SL1 and the first tip **238a** is considered 100% of the first span length SL1. The first span length SL1 is the maximum distance between the first root **237a** and the first tip **238a** of the first airfoil **230a**.

[0084] A first leading edge protector **240a** extends along the chordwise direction C between a first leading edge **235a** and a first seam **239a** to define a first leading length (denoted “FLL”). The first airfoil **230a** has a first chord length (denoted “FCL”) measured along the chordwise direction C between the first leading edge **235a** and the first trailing edge **236a**.

[0085] A relationship between the first leading length (FLL) and the first chord length (FCL) is denoted herein with a first expression of the APF:

$$APF1 = \frac{FLL}{FCL} \quad (1)$$

[0086] OGVs **182** define a second stage of airfoils **200b** downstream from the first stage of airfoils **200a**. A second airfoil **230b** in the second stage of airfoils **200b** is similar to the previously described airfoil **130**, therefore like parts of the second airfoil **230b** will be identified with like numerals increased by 100 and having a notation “b” with it being understood that the description of the like parts of the airfoil **130** applies to the second airfoil **230b**, except where noted. The second airfoil **230b** is located downstream from the first airfoil **230a**. While only a single outlet guide vane **182** is shown in the cross-section it will be understood that the OGVs **182** are multiple OGVs spaced about the fan section **118**.

[0087] A second leading edge protector **240b** extends along the chordwise direction C between a second leading edge **235b** and a second seam **239b** to define a second leading length (denoted “SLL”). The second airfoil **230b** has a second chord length (denoted “SCL”) measured along the chordwise direction C between the second leading edge **235b** and second trailing edge **236b**.

[0088] The second airfoil **230b** has a second span length (denoted “SL2”) measured along the spanwise direction S between a second root **237b** and a second tip **238b** where the second root **237b** is considered 0% of the second span length SL2 and the second tip **238b** is considered 100% of the second span length SL2. The second span length SL2 is the maximum distance between the second root **237b** and the second tip **238b** of the second airfoil **230b**.

[0089] The first and second leading edge protectors **240a**, **240b** can each define first and second sheaths **244a**, **244b**. An exterior surface of each airfoil **230a**, **230b** is defined by the corresponding leading edge protectors **240a**, **240b** and a corresponding composite portion **250a**, **250b**. The composite portions **250a**, **250b** can each include a corresponding

composite leading edge **252a**, **252b** which can define at least a portion of, or all of the corresponding seams **239a**, **239b**.

[0090] A relationship between the second leading length (SLL) and the second chord length (SCL) is denoted herein with a second expression of the APF:

$$APF2 = \frac{SLL}{SCL} \quad (2)$$

[0091] As will be further discussed herein, the APF describes an amount of protection coverage by the leading edge protector of any of the airfoils **130**, **230a**, **230b** described herein. A balance trade-off between the amount of protection and the weight gain/loss associated with any of the protector portions described herein can be expressed by an APF value of from 0.1 to 0.3, inclusive of endpoints. In other words, to satisfy protection requirements the leading edge protector described herein should protect at least 10% and up to and including 30% of the composite airfoil before becoming too heavy.

[0092] The first stage of airfoils **200a** has a first number of airfoils and the second stage of composite airfoils **200b** has a second number of airfoils different than the first number. In other words, the consecutive stages of airfoils can vary in size and number of airfoils. Further, the first stage of composite airfoils **200a** and the second stage of composite airfoils **200b** can both be configured to rotate.

[0093] It will be appreciated that the number, size, and configuration of the composite airfoils described herein are provided by way of example only and that in other exemplary embodiments, the composite airfoils may have any other suitable configuration including that the plurality of airfoils may be in multiple rotor stages, etc.

[0094] As described earlier, finding a workable solution that balances the amount of protective covering for the composite airfoil as described herein whilst maintaining a weight requirement is a labor-intensive and time-intensive process, because the process is iterative and involves the selection of multiple composite airfoils with various protector edge lengths and chord lengths. Design procedures require placing said composite airfoil **130** (FIG. 2) into a turbine engine designed for a first flight operating condition and embodying a protection effectiveness with acceptable weight gain/losses for that first flight operating condition. Evaluating whether in a second, third, or other flight operating condition, the same selected composite airfoil **130** maintains a heat effectiveness with acceptable protection effectiveness for the other operating conditions is time-intensive and necessitates re-design of the composite airfoil and even the turbine engine in the event the conditions are not met. It is desirable to have an ability to arrive at an optimal composite airfoil, like the composite airfoil(s) described herein, rather than relying on chance. It would be desirable to have a limited or narrowed range of possible composite airfoil configurations for satisfying mission requirements, such requirements including protection, weight restrictions, heat transfer, pressure ratio, and noise transmission level requirements, as well as the ability to survive bird strikes at the time a composite airfoil **130** is selected and located within an engine.

[0095] The inventor(s) sought to find the trade-off balance between leading edge protection and weight gain/loss while satisfying all design requirements, because this would yield

a more desired composite airfoil suited for specific needs of the engine, as described above. Knowing these trade-offs is also a desirable time saver.

[0096] TABLE 1 below illustrates some composite airfoil configurations that yielded workable solutions to the trade-off balance problem.

TABLE 1

Example:						
	1	2	3	4	5	6
CL (cm)	47	11	29	60	9.7	13
LL (cm)	11	1.7	3.2	16	1.5	2.3
SL (%)	20	20	38	50	50	80

[0097] It was discovered, unexpectedly, during the course of engine design and the time-consuming iterative process previously described, that a relationship exists between the ratio of the leading length LL to the chord length CL. It has been found that the optimal amount of protective covering of the composite airfoil lies within a specific range based on the leading length LL of the protective covering and the chord length CL of the composite airfoil.

[0098] TABLE 2 below illustrates some consecutive composite airfoil stages with workable solutions to the trade-off balance problem. Different span percentages are shown in TABLE 2. It was found that the CL and LL should be taken for any position between 20% and 80%, inclusive of end points of the span length SL. The specific range of the span length was chosen because the airfoil may have different properties, profiles, etc. at its distal ends. In the non-limiting examples, the fan blade dimensions determine APF1 while the outlet guide vane dimensions determine APF2.

TABLE 2

Fan Blade			Outlet Guide Vane		
Span (%)	CL (cm)	LL (cm)	Span (%)	CL (cm)	LL (cm)
20	46.9	11.2	20	31.4	3.18
24	48.3	11.6	26	30.6	3.18
28	50.5	13.6	32	30.0	3.18
32	52.4	14.2	38	29.3	3.18
36	54.5	14.6	44	28.7	3.18
40	56.5	15.0	50	28.1	3.18
44	58.2	15.3	56	27.5	3.18
48	59.4	15.5	62	26.9	3.18
52	60.1	15.7	68	26.6	3.18
56	60.6	15.6	74	26.7	3.18
60	61.0	15.7	80	27.4	3.18
64	61.5	15.5			
68	61.9	15.4			
72	65.0	15.4			
76	63.2	15.5			
80	64.4	15.7			

[0099] Moreover, utilizing this relationship, the inventors found that the number of suitable or feasible composite airfoil possibilities for placement in a turbine engine that are capable of meeting the design requirements could be greatly reduced, thereby facilitating a more rapid down-selection of composite airfoils to consider as an engine is being developed. Such benefit provides more insight to the requirements for a given engine, and to the requirements for particular composite airfoil locations within the engine, long before specific technologies, integration, or system requirements

are developed fully. The discovered relationship also avoids or prevents late-stage redesign while also providing the composite airfoil with a required protection effectiveness within given weight parameters.

[0100] More specifically, the inventors found that a relationship between the first expression of the APF, APF1, and the second expression of the APF, APF2, optimizes the protection amount for successive stages of airfoils. This relationship was an unexpected discovery during the course of engine design—i.e., designing multistage airfoil sections such as by way of non-limiting examples fan sections, fan blades, and outlet guide vanes and evaluating the impact that an amount of protection on the fan blade has on a needed amount of protection on the outlet guide vane, or vice versa.

[0101] Narrowing the options down based on surrounding stages of airfoils can significantly decrease both material and time costs.

[0102] In other words, an amount of protection provided by the first leading edge protector 240a on the first airfoil 230a can affect an amount of protection necessary for the second airfoil 230b downstream of the first airfoil 230a. This relationship between the multistage airfoils or successive airfoils, such as 230a and 230b, can be described by a stage performance factor (denoted “SPF”) determined from a relationship between the APF1 and the APF2. The stage performance factor can generally be represented by a ratio of the first airfoil protection factor APF1 to the second airfoil protection factor APF2 represented by:

$$SPF = \frac{APF1}{APF2} \quad (3)$$

[0103] More specifically, it was found that for any position between 20% and 80%, inclusive of end points of the span length SL, a desired SPF value is greater than or equal to 0.70 and less than or equal to 4 ($0.7 \leq SPF \leq 4$). The specific range of the span length was chosen because the airfoil may have different properties, profiles, etc. at its distal ends. Conversely, at any position between 20% and 80%, inclusive of end points the airfoil is more uniform and therefore the determined ratios are applicable. It will be understood that because of its position and movement, the rotating fan blade will likely require more coverage from the leading edge protector as compared to a static airfoil or OGV, which is driving the relationship ratio to the 0.7 to 4.0 range. This is due to the fact that the rotating blade has a higher kinetic energy from impact and is driven by the rotating velocity of the airfoil.

[0104] Utilizing this relationship, the inventors were able to arrive at a better performing airfoil in terms of protection amount with acceptable weight increase. The inventors found that the SPF for a set first set of airfoils and a second set of airfoils downstream from the first set of airfoils could be narrowed to an SPF range of greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$). Narrowing the SPF range provides more insight to the requirements for a given engine well before specific technologies, integration and system requirements are developed fully. For example, as the fan speed is reduced, coverage on the first leading edge 235a by the first leading edge protector can decrease such that the APF1 also decreases. Further, knowing a range for the SPF can prevent or minimize late-stage redesign, decrease material cost, and save time.

[0105] The SPF value represents how an amount of protection on a first stage of airfoils, like the first stage of airfoils 200a, impacts an amount of protection necessary for any downstream airfoil stages with respect to the first set of airfoil stages.

[0106] In one example, the set of fan blades 142 illustrated in FIG. 4 can have dimensions of the Fan Blade at 20% from TABLE 2 and the set of outlet guide vanes 182 can have dimensions of the Outlet Guide Vane at 20% from TABLE 2. This results in an APF1 value of (11.2/46.9) or 0.24 and an APF2 value of (3.18/31.4) or 0.10. Using the SPF ratio, an SPF value of (0.24/0.10) or 2.40 is found.

[0107] In another example, the set of fan blades 142 illustrated in FIG. 4 can have dimensions of the Fan Blade at 68% from TABLE 2 and the set of outlet guide vanes 182 can have dimensions of the Outlet Guide Vane at 68% from TABLE 2. This results in an APF1 value of (15.4/61.9) or 0.25 and an APF2 value of (3.18/26.6) or 0.12. Using the SPF ratio, an SPF value of (0.25/0.12) or 2.1 is found.

[0108] Some lower and upper bound values for each design parameter for determining Expression (3) are provided below in TABLE 3:

TABLE 3

Parameter	Lower Bound		Upper Bound	
SL (%)	20	80	20	80
First Airfoil				
FCL (cm)	24	32	56	77
FLL (cm)	6	8	13	19
Second Airfoil				
SCL (cm)	9.9	9.3	31	27
SLL (cm)	1.6	1.5	4	3.5

[0109] It was found that first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 below fit into the composite airfoil dimensions previously described herein. These ranges enable a minimum weight gain for a compact and proficiently protected composite airfoils in succession.

TABLE 4

Ratio	Narrow Range	Broad Range
SPF	0.95-2.5	0.70-4.0
APF1	0.22-0.25	0.20-0.30
APF2	0.10-0.12	0.08-0.17

[0110] Pairs of first and second airfoils, with the second airfoils downstream of the first airfoils within the ranges provided can be assembled to conform with any fan section, or other downstream stage relationship for blades/vanes and blades/blades. This can include any number of engine designs including ducted and unducted engines as well as a direct-drive configuration and an indirect-drive configuration such as a speed reduction device or a geared-drive configuration.

[0111] For example, FIG. 5 illustrates a gas turbine engine 310 as a high-bypass turbofan jet engine, sometimes also referred to as a turbofan engine, which can include the set of composite airfoils or first and second stages of composite airfoils as described herein. The gas turbine engine 310 defines an axial direction A (extending parallel to a longi-

tudinal centerline 312 provided for reference), a radial direction R, and a circumferential direction CD extending about the longitudinal centerline 312. In general, the gas turbine engine 310 includes a fan section 314 and a turbomachine 316 disposed downstream from the fan section 314.

[0112] The exemplary turbomachine 316 depicted generally includes a substantially tubular outer casing 318 that defines an annular inlet 320. The outer casing 318 encases, in serial flow relationship, a compressor section including a booster or low pressure (LP) compressor 322 and a high pressure (HP) compressor 324, a combustion section 326, a turbine section including a high pressure (HP) turbine 328 and a low pressure (LP) turbine 330, and a jet exhaust nozzle section 332. A high pressure (HP) shaft 334, which may additionally or alternatively be a spool, drivingly connects the HP turbine 328 to the HP compressor 324. A low pressure (LP) shaft 336, which may additionally or alternatively be a spool, drivingly connects the LP turbine 330 to the LP compressor 322. The compressor section, combustion section 326, turbine section, and jet exhaust nozzle section 332 together define a working gas flow path 337.

[0113] In the illustrated example, and by way of non-limiting example, the fan section 314 includes a fan 338 having a plurality of fan blades 340 coupled to a disk 342 in a spaced apart manner. As depicted, the fan blades 340 extend outwardly from disk 342 generally along the radial direction R.

[0114] Each fan blade 340 is rotatable relative to the disk 342 about a pitch axis P by virtue of the fan blades 340 being operatively coupled to a suitable pitch change mechanism 344 configured to collectively vary the pitch of the fan blades 340, e.g., in unison. The gas turbine engine 310 further includes a speed reduction device in the form of a power gearbox 346, and the fan blades 340, disk 342, and pitch change mechanism 344 are together rotatable about the longitudinal centerline 312 by LP shaft 336 across the power gearbox 346. The power gearbox 346 includes a plurality of gears for adjusting a rotational speed of the fan 338 relative to a rotational speed of the LP shaft 336, such that the fan 338 may rotate at a more efficient fan speed. It will be understood that any suitable speed reduction device configured to adjust the rotation of the fan 338 relative to the LP shaft 336 can be utilized and that a power gearbox is merely one example thereof.

[0115] The disk 342 is covered by rotatable front hub 348 of the fan section 314. The front hub 348 is also sometimes referred to as a spinner. The front hub 348 is aerodynamically contoured to promote an airflow through the plurality of fan blades 340.

[0116] Additionally, the exemplary fan section 314 includes an annular fan casing or outer nacelle 350 that circumferentially surrounds the fan 338, circumferentially surrounds at least a portion of the turbomachine 316, or a combination thereof. It should be appreciated that the nacelle 350 is supported relative to the turbomachine 316 by a plurality of outlet guide vanes 352, which can be a second stage of airfoils in the non-limiting example. Moreover, a downstream section 354 of the nacelle 350 extends over an outer portion of the turbomachine 316 so as to define a bypass airflow passage 356 therebetween.

[0117] It will be understood that each fan blade of the plurality of fan blades 340 may form a composite airfoil and that the plurality of fan blades 340 can form a first stage of

airfoils as described above. More specifically, each of the plurality of fan blades **340** can include a first leading edge protector **340a**. It will be understood that the plurality of fan blades forming the first stage of airfoils are similar to the previously described airfoils **130** and **230a** with it being understood that the description of like parts applies to the plurality of fan blades unless otherwise noted.

[0118] Further still, it will be understood that each outlet guide vane of the plurality of outlet guide vanes **352** may form a composite airfoil. Further still, in the illustrated example, the plurality of outlet guide vanes **352** can form a second stage of airfoils as described above. More specifically, each of the plurality of outlet guide vanes **352** can include a second leading edge protector **352a**. It will be understood that an outlet guide vane of the plurality of outlet guide vanes **352** forming the second stage of airfoils is similar to the previously described airfoils **130** and **230b** with it being understood that the description of like parts applies to the outlet guide vane of the plurality of outlet guide vanes **352** unless otherwise noted.

[0119] It will be understood that the plurality of fan blades **340** and the plurality of outlet guide vanes **352** are similar to the previously described first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 above.

[0120] During operation of the gas turbine engine **310**, a volume of air **358** enters the gas turbine engine **310** through an associated inlet **360** of the nacelle **350** and fan section **314**. As the volume of air **358** passes across the fan blades **340**, a first portion of air **362** is directed or routed into the bypass airflow passage **356** and a second portion of air **364** as indicated by arrow **364** is directed or routed into the working gas flow path **337**, or more specifically into the LP compressor **322**. The ratio between the first portion of air **362** and the second portion of air **364** is commonly known as a bypass ratio. A pressure of the second portion of air **364** is then increased as it is routed through the HP compressor **324** and into the combustion section **326**, where it is mixed with fuel and burned to provide combustion gases **366**.

[0121] The combustion gases **366** are routed through the HP turbine **328** where a portion of thermal and/or kinetic energy from the combustion gases **366** is extracted via sequential stages of HP turbine stator vanes **368** that are coupled to the outer casing **318** and HP turbine rotor blades **370** that are coupled to the HP shaft **334**, thus causing the HP shaft **334** to rotate, which supports operation of the HP compressor **324**. The combustion gases **366** are then routed through the LP turbine **330** where a second portion of thermal and kinetic energy is extracted from the combustion gases **366** via sequential stages of LP turbine stator vanes **372** that are coupled to the outer casing **318** and LP turbine rotor blades **374** that are coupled to the LP shaft **336**, thus causing the LP shaft **336** to rotate, which supports operation of the LP compressor **322**, rotation of the fan **338**, or a combination thereof.

[0122] The combustion gases **366** are subsequently routed through the jet exhaust nozzle section **332** of the turbomachine **316** to provide propulsive thrust. Simultaneously, the pressure of the first portion of air **362** is substantially increased as the first portion of air **362** is routed through the bypass airflow passage **356** before it is exhausted from a fan nozzle exhaust section **376** of the gas turbine engine **310**, also providing propulsive thrust. The HP turbine **328**, the LP turbine **330**, and the jet exhaust nozzle section **332** at least

partially define a hot gas path **378** for routing the combustion gases **366** through the turbomachine **316**.

[0123] As previously described the stages of airfoils exemplary gas turbine engine **310** depicted in FIG. 5 is by way of example only, and that in other exemplary embodiments, the gas turbine engine **310** may have other configurations. For example, although the gas turbine engine **310** depicted is configured as a ducted gas turbine engine (i.e., including the outer nacelle **350**, also referred to herein as a turbofan engine), in other embodiments, the gas turbine engine **310** may be an unducted gas turbine engine (such that the fan **338** is an unducted fan, and the outlet guide vanes **352** are cantilevered from the outer casing **318**; see, e.g., FIG. 6; also referred to herein as an open rotor engine). Additionally, or alternatively, although the gas turbine engine **310** depicted is configured as a variable pitch gas turbine engine (i.e., including a fan **338** configured as a variable pitch fan), in other embodiments, the gas turbine engine **310** may alternatively be configured as a fixed pitch gas turbine engine (such that the fan **338** includes fan blades **340** that are not rotatable about a pitch axis P).

[0124] FIG. 6, illustrates another non-limiting example of a gas turbine engine **400**, which can include the set of composite airfoils or first and second stages of composite airfoils as described herein. The exemplary gas turbine engine **400** of FIG. 4 may be configured in substantially the same manner as the exemplary gas turbine engine **310** described above with reference to FIG. 5.

[0125] For example, the exemplary gas turbine engine **400** defines an axial direction A, a radial direction R, and a circumferential direction CD. Moreover, the engine **400** defines an axial centerline, longitudinal axis or engine centerline **412** that extends along the axial direction A. In general, the axial direction A extends parallel to the engine centerline **412**, the radial direction R extends outward from and inward to the engine centerline **412** in a direction orthogonal to the axial direction A, and the circumferential direction CD extends three hundred sixty degrees (360°) around the engine centerline **412**. The engine **400** extends between a forward end **414** and an aft end **416**, e.g., along the axial direction A.

[0126] Further, the exemplary gas turbine engine **400** generally includes a fan section **450** and a turbomachine **420**. Generally, the turbomachine **420** includes, in serial flow order, a compressor section, a combustion section, a turbine section, and an exhaust section. In a non-limiting example, the turbomachine **420** includes a core cowl **422** that defines a core inlet **424** that is annular. The core cowl **422** further encloses at least in part a low pressure system and a high pressure system. For example, the core cowl **422** depicted encloses and supports at least in part a booster or low pressure ("LP") compressor **426**, a high pressure ("HP") compressor **428**, a combustor **430**, a high pressure turbine **432**, and a low pressure turbine **434**. The high pressure turbine **432** drives the high pressure compressor **428** through a high pressure shaft **436**. The low pressure turbine **434** drives the low pressure compressor **426** and components of the fan section **450** through a low pressure shaft **438**. After driving each of the high pressure turbine **432** and the low pressure turbine **434**, combustion products exit the turbomachine **420** through a turbomachine exhaust nozzle **440**.

[0127] In this manner, the turbomachine **420** defines a working gas flow path or core duct **442** that extends between the core inlet **424** and the turbomachine exhaust nozzle **440**.

The core duct **442** is an annular duct positioned generally inward of the core cowl **422** along the radial direction R. The core duct **442** may be referred to as a second stream.

[0128] The fan section **450** includes a fan **452**, which is the primary fan in non-limiting example. One difference is that the fan **452** is an open rotor or unducted fan. In such a manner, the gas turbine engine **400** may be referred to as an open rotor engine. The fan **452** includes fan blades **454**, while only a single fan blade is illustrated in FIG. 6 it will be understood that an array of fan blades are included. Moreover, the fan blades **454** can be arranged in equal spacing around the engine centerline **412**. Each fan blade **454** has a root and a tip and a span defined therebetween. Each fan blade **454** defines a central blade axis **456**. For this embodiment, each fan blade **454** of the fan **452** is rotatable about its central blade axis **456**, e.g., in unison with one another. One or more actuators **458** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan blades **454**.

[0129] The fan blades **454** are rotatable about the engine centerline **412**. As noted above, the fan **452** is drivingly coupled with the low pressure turbine **434** via the LP shaft **438**. In a non-limiting example, the fan **452** is coupled with the LP shaft **438** via a speed reduction device, which can include by way of non-limiting examples a power gearbox or a speed reduction gearbox **455**, e.g., in an indirect-drive or geared-drive configuration.

[0130] The fan section **450** further includes a fan guide vane array **460** that includes fan guide vanes **462**, again while only one fan guide vane is shown in FIG. 6 it will be understood that the fan guide vanes **462** are disposed around the engine centerline **412**. The fan guide vanes **462** are mounted to the fan cowl **470**. In a non-limiting example, the fan guide vanes **462** are not rotatable about the engine centerline **412**. Each of the fan guide vanes **462** has a root and a tip and a span defined therebetween. The fan guide vanes **462** may be unshrouded as shown in FIG. 6 or, alternatively, may be shrouded, e.g., by an annular shroud spaced outward from the tips of the fan guide vanes **462** along the radial direction R or attached to the fan guide vanes **462**.

[0131] Each fan guide vane **462** defines a central blade axis **464**. By way of non-limiting example, each of the fan guide vanes **462** of the fan guide vane array **460** is rotatable about its respective central blade axis **464**, e.g., in unison with one another. One or more actuators **466** are provided to facilitate such rotation and therefore may be used to change a pitch of the fan guide vane **462** about its respective central blade axis **464**. However, in other embodiments, each of the fan guide vanes **462** may be fixed or unable to be pitched about its central blade axis **464**.

[0132] It will be understood that each of the fan blades **454** may form a composite airfoil and that the fan blades **454** can form a first stage of airfoils as described above. More specifically, each of the fan blades **454** can include a first leading edge protector **454a**. It will be understood that the fan blades **454** forming the first stage of airfoils are similar to the previously described airfoils **130**, **230a**, and **340** with it being understood that the description of like parts applies to the fan blades unless otherwise noted.

[0133] Further still, it will be understood that each of the fan guide vanes **462** may form a composite airfoil. Further still, in the illustrated example, the fan guide vanes **462** can form a second stage of airfoils as described above. More

specifically, each of the fan guide vanes **462** can include a second leading edge protector **462a**. It will be understood that the fan guide vanes **462** forming the second stage of airfoils is similar to the previously described airfoils **130**, **230b**, and **352** with it being understood that the description of like parts applies to the fan guide vanes **462** unless otherwise noted.

[0134] It will be understood that the fan blades **454** and the fan guide vanes **462** are similar to the previously described first and second airfoil pairs with dimensions fitting in the ranges set out in TABLE 4 above.

[0135] Another difference is that the illustrated example in FIG. 6, in addition to the unducted fan **452**, shows a ducted fan **484** included aft of the fan **452**. In this manner, the engine **400** includes both a ducted fan **484** and an unducted fan **452**, which both serve to generate thrust through the movement of air without passage through at least a portion of the turbomachine **420** (e.g., without passage through the HP compressor **428** and combustion section for the embodiment depicted). The ducted fan **484** is rotatable about the engine centerline **412**. The ducted fan **484** is, by way of non-limiting example, driven by the low pressure turbine **434** (e.g. coupled to the LP shaft **438**). The fan **452** may be referred to as the primary fan, and the ducted fan **484** may be referred to as a secondary fan. It will be appreciated that these terms “primary” and “secondary” are terms of convenience, and do not imply any particular importance, power, or the like.

[0136] The ducted fan **484** includes a plurality of fan blades (not separately labeled in FIG. 6) arranged in a single stage, such that the ducted fan **484** may be referred to as a single stage fan. The fan blades of the ducted fan **484** can be arranged in equal spacing around the engine centerline **412**. Each blade of the ducted fan **484** has a root and a tip and a span defined therebetween.

[0137] The fan cowl **470** annularly encases at least a portion of the core cowl **422** and is generally positioned outward of at least a portion of the core cowl **422** along the radial direction R. Particularly, a downstream section of the fan cowl **470** extends over a forward portion of the core cowl **422** to define a fan duct flowpath, or simply a fan duct **472**. The fan flowpath or fan duct **472** may be understood as forming at least a portion of the third stream of the engine **400**.

[0138] Incoming air may enter through the fan duct **472** through a fan duct inlet **476** and may exit through a fan exhaust nozzle **478** to produce propulsive thrust. The fan duct **472** is an annular duct positioned generally outward of the core duct **442** along the radial direction R. The fan cowl **470** and the core cowl **422** are connected together and supported by a plurality of substantially radially extending, circumferentially-spaced stationary struts **474** (only one of which is shown in FIG. 6). The stationary struts **474** may each be aerodynamically contoured to direct air flowing thereby. Other struts in addition to the stationary struts **474** may be used to connect and support the fan cowl **470**, the core cowl **422**, or a combination thereof. In many embodiments, the fan duct **472** and the core duct **442** may at least partially co-extend axially on opposite radial sides of the core cowl **422**. For example, the fan duct **472** and the core duct **442** may each extend directly from a leading edge **444** of the core cowl **422** and may partially co-extend generally axially on opposite radial sides of the core cowl **422**.

[0139] The engine 400 also defines or includes an inlet duct 480. The inlet duct 480 extends between the engine inlet 482 and the core inlet 424, the fan duct inlet 476, or a combination thereof. The engine inlet 482 is defined generally at the forward end of the fan cowl 470 and is positioned between the fan 452 and the fan guide vane array 460 along the axial direction A. The inlet duct 480 is an annular duct that is positioned inward of the fan cowl 470 along the radial direction R. Air flowing downstream along the inlet duct 480 is split, not necessarily evenly, into the core duct 442 and the fan duct 472 by a fan duct splitter or leading edge 444 of the core cowl 422. In the embodiment depicted, the inlet duct 480 is wider than the core duct 442 along the radial direction R. The inlet duct 480 is also wider than the fan duct 472 along the radial direction R.

[0140] Air passing through the fan duct 472 may be relatively cooler than one or more fluids utilized in the turbomachine 420. In this way, one or more heat exchangers 486 may be positioned in thermal communication with the fan duct 472. For example, one or more heat exchangers 486 may be disposed within the fan duct 472 and utilized to cool one or more fluids from the core engine with the air passing through the fan duct 472, as a resource for removing heat from a fluid, e.g., compressor bleed air, oil or fuel. The heat exchanger 486 may be an annular heat exchanger.

[0141] In various embodiments, it will be appreciated that the engine includes a ratio of a quantity of vanes to a quantity of blades that could be less than, equal to, or greater than 1:1. For example, in particular embodiments, the engine includes twelve (12) fan blades and ten (10) vanes. In other embodiments, the vane assembly includes a greater quantity of vanes to fan blades. For example, in particular embodiments, the engine includes ten (10) fan blades and twenty-three (23) vanes. For example, in certain embodiments, the engine may include a ratio of a quantity of vanes to a quantity of blades between 1:2 and 5:2. The ratio may be tuned based on a variety of factors including a size of the vanes to ensure a desired amount of swirl is removed for an airflow from the primary fan.

[0142] It should be appreciated that various embodiments of the engine, such as the single unducted rotor engine depicted and described herein, may allow for normal subsonic aircraft cruise altitude operation at or above Mach 0.5. In certain embodiments, the engine allows for normal aircraft operation between Mach 0.55 and Mach 0.85 at cruise altitude. In still particular embodiments, the engine allows for normal aircraft operation between Mach 0.75 and Mach 0.85. In certain embodiments, the engine allows for rotor blade tip speeds at or less than 750 feet per second (fps). In other embodiments, the rotor blade tip speed at a cruise flight condition can be 650 to 900 fps, or 700 to 800 fps.

[0143] It will be understood that a speed reduction device including, but not limited to, a gear assembly may be provided to reduce a rotational speed of the fan assembly relative to a driving shaft (such as a low pressure shaft coupled to a low pressure turbine). In some embodiments, a gear ratio of the input rotational speed to the output rotational speed is greater than or equal to 2. For example, in particular embodiments, the gear ratio is within a range of 4.1 to 14.0, within a range of 4.5 to 14.0, or within a range of 6.0 to 14.0. In certain embodiments, the gear ratio is within a range of 4.5 to 12 or within a range of 6.0 to 11.0. As such, in some embodiments, the fan can be configured to rotate at a rotational speed of 700 to 1500 revolutions per

minute (rpm) at a cruise flight condition, while the power turbine (e.g., the low-pressure turbine) is configured to rotate at a rotational speed of 2,500 to 15,000 rpm at a cruise flight condition. In particular embodiments, the fan can be configured to rotate at a rotational speed of 850 to 1,350 rpm at a cruise flight condition, while the power turbine is configured to rotate at a rotational speed of 5,000 to 10,000 rpm at a cruise flight condition.

[0144] With respect to a turbomachine of the gas turbine engine, the compressors and/or turbines can include various stage counts. As disclosed herein, the stage count includes the number of rotors or blade stages in a particular component (e.g., a compressor or turbine). For example, in some embodiments, a low pressure compressor may include 1 to 8 stages, a high-pressure compressor may include 8 to 15 stages, a high-pressure turbine may include 1 to 2 stages, and/or a low pressure turbine (LPT) may include 3 to 7 stages. In particular, the LPT may have 4 stages, or between 4 and 7 stages. For example, in certain embodiments, an engine may include a one stage low pressure compressor, an 11 stage high pressure compressor, a two stage high pressure turbine, and 4 stages, or between 4 and 7 stages for the LPT. As another example, an engine can include a three stage low-pressure compressor, a 10 stage high pressure compressor, a two stage high pressure turbine, and a 7 stage low pressure turbine.

[0145] The SPF is useful for making trade-offs when determining an amount of protection on a first airfoil in relationship to an amount on an airfoil downstream of the first airfoil. For example, when there is a limited space available for a fan blade in a fan section, knowledge of those dimensions and the downstream airfoil dimensions enables determination of an acceptable cover with a leading edge protector length allowing for sufficient leading edge protection.

[0146] Benefits associated with the SPF described herein include a quick assessment of design parameters in terms of composite airfoils in downstream relationship. Further, the SPF described herein enables a quick visualization of tradeoffs in terms of geometry that are bounded by the constraints imposed by the materials used, the available space in which the composite airfoils are located, the type of turbine engine or system enclosures and the configuration of surrounding components, or any other design constraint. The SPF enables the manufacturing of a high performing composite airfoil with peak performance with the factors available. While narrowing these multiple factors to a region of possibilities saves time, money, and resources, the largest benefit is at the system level, where the composite airfoils described herein enable improved system performance. Previously developed composite airfoils may peak in one area of performance by design, but lose efficiency or lifetime benefits in another area of performance. In other words, the stage performance factor enables the development and production of higher performing composite airfoils across multiple performance metrics within a given set of constraints.

[0147] As discussed, the inventors have developed airfoils, specifically composite fan blades and composite outlet guide vanes, that balance an added weight component from a protective covering on the leading edge with an acceptable amount of protection of the leading edge. The inventors have further sought to balance this weight-protection relationship while also maximizing the aerodynamic loading of the blades.

[0148] More particularly, the inventors have unexpectedly discovered that the above-described composite blades having a protective covering on the leading edge demonstrate improved aerodynamic loading when additionally implemented with stagger and camber considerations between a blade root and a blade tip, along with chord length and sweep considerations.

[0149] As will be further appreciated, the inventors have discovered a fan blade having a combination of aerodynamic sweep in the leading edge and locally increased camber in the outer panel that is effective for increasing hub supercharging of the fan air while maintaining aeromechanical stability. The increased camber of the airfoil outer panel is further effective for delaying flutter and tailors the efficiency characteristics of the fan for improvement, particularly at part speed corresponding with aircraft cruise operation. Maintaining the aerodynamic sweep near the airfoil hub or root at a relatively low level in the aft direction may be used for minimizing the required twist in the airfoil, which in turn will moderate or reduce the stresses in the blade root and dovetail, which is beneficial for the composite blades discussed herein.

[0150] As a consequence of the moderate aft sweep in the airfoil hub region, the tendency for radial outflow of the air being pressurized will be reduced for providing more turning in the air from a given camber level. Increased camber in the outer panel of the airfoil correspondingly reduces the incidence angle of the incoming air and correspondingly improves aeromechanical characteristics in the part speed region of the operating envelope.

[0151] Aerodynamic efficiency may be further improved in the turbofan engine 10 illustrated in FIG. 1 by reducing the solidity at the airfoil tips by reducing the number of blades (e.g., from twenty-two to twenty or eighteen) while maintaining a similar ratio of the tip chord over the tip diameter C/D in the fan.

[0152] In this regard, and referring to FIG. 7, adjacent airfoils 500 define circumferentially therebetween corresponding flow passages 552 for pressurizing air 504 during operation. Each of the airfoils 500 includes stagger (or twist) represented by the stagger angle A from the axial or longitudinal axis.

[0153] The stagger increases from the root to the tip 526 of the airfoil 500. For example, the stagger angle A at the blade tip 526 may be relatively substantial (e.g., about 60 degrees) to position the leading edge 510 of one airfoil circumferentially adjacent but axially spaced from a suction side 524 of the next adjacent airfoil aft from the leading edge thereof to define a corresponding mouth 554 for the flow passage between the opposing pressure and suction sides 522, 524 of the adjacent airfoils 500. The contours and stagger of the adjacent airfoils 500 over the radial span of the blades cause each flow passage 552 to converge or decrease in flow area to a throat 556 of minimum flow area spaced aft from the mouth 554 along most, if not all, of the radial span.

[0154] As further illustrated in FIG. 7, a relatively high airfoil stagger A also positions the trailing edge 512 of one airfoil 500 circumferentially adjacent to the pressure side 522 of the next adjacent airfoil 500 while also being spaced axially therefrom in the tip region to define a corresponding discharge or outlet 558 for the corresponding flow passage 552 between adjacent airfoils 500. In this way, the incoming air 504 is channeled in the corresponding flow passages 552

between adjacent airfoils 500 as they rotate clockwise in FIG. 7 for pressurizing the air 504 to produce the propulsion thrust during operation.

[0155] As shown in FIG. 7, a sloped platform 560 is disposed between corresponding pairs of adjacent airfoils 500 to conform with or match roots 502 of the blades, which may be in the form of sloped roots 502 to define the radially inner boundary of the fan air flowpath. Each platform 560 may be a discrete or separate component suitably mounted to the supporting fan disk between adjacent airfoils, or may be formed in halves integrally formed with the opposite sides of each fan blade along the roots thereof. The platform 560 slopes radially outwardly in the downstream or aft direction between the leading and trailing edges of the blade at a slope angle B (shown in FIG. 9) of about 15-20 degrees, and defines the radially inner boundary of each flow passage 552 between the blades. The sloped platform 560 cooperates with the airfoils themselves in pressurizing the airflow flowing downstream therepast.

[0156] The radially inner portion of an airfoil 500 typically operates with subsonic airflow thereover for supercharging or initially pressurizing the inner portion of the air 504 as it enters a booster compressor. The outer portion of the fan blade has the outer tip 526 disposed at a suitably large outer diameter and rotates at a corresponding rotational speed for effecting supersonic airflow therepast during certain portions of the operating envelope of the aircraft being powered by the engine.

[0157] Another feature of the airfoil 500 that affects its aerodynamic performance is its camber, which represents the amount of curvature of the airfoil along the radial or transverse sections between the leading and trailing edges 510, 512 thereof. FIG. 8 illustrates in solid line two graphs of exemplary camber and slope thereof over the span of the airfoil 500 from its root 502 at zero percent span to its tip 526 at 100 percent span, along with two representative radial sections of the airfoil at 70 percent and 100 percent span.

[0158] The airfoil camber may be defined by the difference in local axial inlet angle E at the leading edge 510 and the local axial exit angle F at the trailing edge 512. The camber line of the airfoil 500 is the mean line extending between the leading and trailing edges 510, 512 of each section between the opposite pressure and suction sides 522, 524. At the leading edge 510, the camber line defines the inlet angle E relative to the axial or longitudinal axis of the engine, and at the trailing edge 512, the camber line defines the exit angle F also relative to the axial axis.

[0159] The camber for each radial section of the airfoil 500 may be simply obtained by subtracting the exit angle F from the inlet angle E, with the resulting camber represented in degrees as shown in the camber graph from root 502 to tip 526 of the airfoil 500. The graph illustrates that the camber for a fan blade decreases between the root 502 and tip 526 of the airfoil 500 in a substantially smooth curve from root to just short of the tip 526, with a slight local increase or peak in camber in the outer span of the airfoil 500 immediately below the tip 526.

[0160] As indicated above, the airfoil 500 illustrated in FIG. 7 has substantial twist or stagger A of about 60 degrees for example from root 502 to tip 526, along with a corresponding variation in camber of the individual radial sections from root 502 to tip 526, and with a relatively large chord. The airfoil 500 is specifically configured to distinctly include sharp leading and trailing edges 510, 512 from root

502 to tip **526** which gradually increase in airfoil thickness to a maximum thickness along the midchord region between the opposite leading and trailing edges **510**, **512**.

[0161] For example, the sharp leading edge **510** of the airfoil **500** may be defined by a circumscribed square outline in the exemplary range of 26 to 42 mils (0.66-1.0 mm) thickness, with the sharp trailing edge **512** being represented by an inscribed circle of diameter 28 to 40 mils (0.7-1.0 mm). The maximum thickness of the airfoil **500** correspondingly ranges from about 120 to 260 mils (3-6.6 mm) from root **502** to tip **526**, which maximum thickness varies along the chord in the exemplary range of 40 percent to 60 percent from the leading edge **510** in axial projection of the twisted airfoil **500** along the longitudinal engine axis.

[0162] As shown in FIG. 7, the stagger A and length of the chord C combine to effect the three-dimensional (3D) configuration of each airfoil **500**. The section chords of the airfoil **500** typically increase in length outboard from the root **502** to correspondingly barrel the airfoil **500** above the root **502**. The airfoil or chord barreling may be observed in the axial side projection, which locally enlarges the midspan region of the airfoil **500** preferably along both the leading and trailing edges **510**, **512**. The maximum airfoil barrel occurs at a suitable midspan region of the airfoil at an intermediate radial section of about 40 percent span from the root **502** in the exemplary embodiment illustrated.

[0163] The leading edge barreling extends in axial projection the leading edge upstream or forward of a straight line extending between the root **502** and tip **526** at the leading edge **510**, and correspondingly the trailing edge barreling extends in axial projection the trailing edge downstream or aft of a straight line extending between the root **502** and tip **526** at the trailing edge **512**. In this way, the leading edge **510** in the barrel extends axially forward of the root **502**, and the trailing edge **512** is correspondingly barreled and also extends axially aft from the root **502**.

[0164] The airfoil barreling is yet another feature of the airfoil **500** which affects the 3D configuration thereof and its aerodynamic performance in pressurizing the airflow channeled thereover during operation. In this regard, aerodynamic sweep is a conventional parameter represented by a local sweep angle, which is a function of the direction of the incoming air and the orientation of the airfoil surface in both the axial and circumferential, or tangential, directions.

[0165] FIG. 9 illustrates the airfoil **500** along with the preferred aerodynamic sweep angle thereof as represented by the upper case letter S which has a negative value (−) for forward sweep, and a positive value (+) for aft sweep. A graph of aerodynamic sweep S for the leading edge **510** from root **502** at zero percent span to tip **526** at 100 percent span is superimposed over the airfoil **500**.

[0166] The airfoil **500** preferably includes forward aerodynamic sweep S-at the tip **526** from leading edge **510** to trailing edge **512**. Chord barreling of the airfoil **500** in conjunction with the forward tip sweep has significant aerodynamic benefits including increased flow capacity at high or maximum fan speed, while also improving part speed efficiency and stability margin.

[0167] However, the airfoil **500** illustrated in FIG. 9 preferably includes non-forward aerodynamic sweep between the airfoil maximum barrel radial section and the sloped root **502** for further improving performance of the

airfoil **500**, particularly for part speed operability favoring cruise operation of the engine in propelling the aircraft in flight.

[0168] In the preferred embodiment illustrated in FIG. 9, the airfoil **500** includes relatively low aerodynamic sweep in the hub region near the root **502** or platform **560** which is less than about half the maximum sweep in the airfoil **500** thereabove.

[0169] It is noted that the radial span of the airfoil **500** is relatively large in the turbine engine **10** illustrated in FIG. 1, with the inner portion or panel of the airfoil **500** below the midspan operating with relatively low Mach velocity of the airflow as it is pressurized or supercharged prior to entry in the compressor section **22**. The radially outer portion or panel of each airfoil **500** above the midspan is highly twisted and is operated in transonic to supersonic Mach velocity of the airflow for providing a substantial pressure increase in the fan air used for generating the substantial amount of propulsion thrust from the turbine engine **10**.

[0170] The relatively low level of aerodynamic sweep which is preferably aft sweep near the airfoil root **502** is found in the low Mach number region of the blade and serves to moderate the twist and leading edge stress in the root **502** for mechanical benefits and to improve the pressurizing capability for a given level of camber.

[0171] FIG. 9 illustrates that the initially forward sweep S− at the airfoil tip **526** transitions through zero sweep to aft aerodynamic sweep S+ inwardly therefrom in the airfoil barrel both along the leading and trailing edges **510**, **512**. In particular, the airfoil has maximum forward sweep along the leading edge **510** at the tip **526** which transitions to maximum aft sweep inwardly therefrom along the leading edge **510** in the barrel region of the airfoil **500**. From the maximum aft sweep, the sweep decreases in aft magnitude along the leading edge **510** towards the root **502** at zero span.

[0172] In the embodiment illustrated in FIG. 9, the airfoil **500** includes zero sweep along the leading edge **510** at the root **502**, along with a local smaller peak in aft sweep along the leading edge **510** between the root **502** and the maximum aft sweep. The maximum aft sweep along the leading edge **510** occurs at about 60 percent span from the root, with the sweep remaining aft in magnitude over the entire inner panel span from 60 percent down to the root **502** where it returns to zero sweep.

[0173] The leading edge sweep illustrated in FIG. 9 decreases rapidly from the maximum aft sweep near the midspan to the relatively low aft sweep in the inner panel down to the root **502** which is substantially less than the magnitude of the maximum aft sweep. The local peak in leading edge sweep occurs at about five percent span and is less than half the magnitude of the maximum aft sweep, with the aft sweep being substantially lower in magnitude over most of the inner panel to about the 40 percent span elevation.

[0174] As indicated above with respect to FIG. 8, the airfoil camber is defined by the difference in the inlet and exit angles (E-F). The inlet angle E preferably increases smoothly over the span of the airfoil **500** from root **502** to tip **526**, with the exit angle F of the airfoil increasing from root **502** to tip **526** at a greater rate with about twice the overall magnitude. The increase in exit angle from root **502** to tip **526** may be used to correspondingly decrease the camber from the root **502** toward the tip **526**, with the camber graph in FIG. 8 illustrating a local increase in

camber in the outer panel immediately below the airfoil tip **526** down to about the midspan within the barrel.

[0175] FIG. 8 further illustrates that the camber has a greater rate of decrease, or slope, in the inner panel radially outboard from the root **502** into the barrel to about 50 percent span for example than in the outer panel outboard from within the barrel at the 50 percent span for example toward the radially outer tip **526**, which correspondingly locally increases the camber near or just below the airfoil tip **526** over most of the outer panel.

[0176] The slope graph of camber illustrated in FIG. 8 shows the pronounced difference in a derived fan blade or airfoil **500** over a preexisting blade **562** for introducing the local increase or peak in camber in the outer panel just below the airfoil tip **526**. The slope represents the change in camber over the radial span of the airfoil from root **502** to tip **526**.

[0177] The slope in both curves is initially negative from the root **502** outward as the camber decreases in magnitude outwardly along the span. The decreasing slope is similar in the two curves in the lower panel of the airfoils **500**, and is affected in large part by the specific blade count in the full complement of fan blades in the fan row.

[0178] However, from the midspan region outwardly above the barrel in the outer panel, the camber and slope thereof in the solid and dashed curves change significantly. In the dashed curves, the camber decreases outwardly over the inner panel, and then increases locally over the outer panel. The slope or rate of camber change has a negative peak at 20% span in the inner panel, and transitions to a positive peak at the airfoil tip **526** after passing through the zero value at about 73% span. This corresponds with a local minimum peak in camber for the preexisting fan blade **562**.

[0179] In contrast, the slope of the camber in the derived fan blade over the preexisting blade, although generally similar over the inner panel, is remarkably different over the outer panel so that the camber has a local increase or peak in magnitude at about 80% span between the tip **526** and barrel.

[0180] The camber as shown by the solid curve varies in slope with a local negative magnitude or peak in the slope immediately below the tip **526** at about 94% span, followed by a local positive magnitude or peak in the slope at about 70% span therebelow. The solid slope curve therefore crosses the zero line at two locations above the midspan at about 62% and 74% span. And, below the midspan, the solid slope curve has a maximum negative peak at about 15% span.

[0181] Accordingly, the camber and its slope in the derived fan blade **500** are specifically configured to effect the local increase in camber in the outer panel just below the tip, with the camber level returning to its minimum value at the airfoil tip **526**. In this way, a significant improvement in fan performance and efficiency may be obtained, while reducing efficiency losses at the airfoil tip itself which has a camber level closely similar to conventional practice.

[0182] The additional camber of the outer panel of the airfoil may reduce high speed performance of the fan, but can be balanced by the improved performance at part speed, specifically cruise operation of the engine for maximizing overall efficiency of the fan.

[0183] Furthermore, the reduction in number of fan blades increases the circumferential pitch P between the airfoils and increases the flow area of the flow passages **552**, in particu-

lar at the throats **556** thereof, for reducing flow blockage during operation, and specifically at the airfoil tips subject to supersonic operation.

[0184] The configuration of the flow passage **552** illustrated in FIG. 7 is particularly important to efficient operation of the fan, and in particular at the airfoil tips subject to supersonic flow. The specific profiles of the pressure and suction sides **522**, **524** of the individual airfoils **500**, the lateral thickness T of the airfoil, the root-to-tip stagger and camber of the airfoils **500** and, of course, the reduced solidity due to the reduction in blade count while maintaining equal the chord to diameter C/D ratio are all used to define each flow passage **552**.

[0185] In particular, the airfoil tips **500** are locally angled and vary in thickness T or width between the leading and trailing edges **510**, **512** to typically converge the flow passage **552** at the airfoil tips from the mouth **554** to the throat **556** and then diverge the flow passage also at the tip from the throat **556** to the outlet **558**. Alternatively, the mouth **554** and throat **556** of the flow passages **552** at the airfoil tips **526** may be coincident in one plane at the leading edges **510**, with the flow passages **552** still diverging aft from the throats **556** at the leading edges **510** to the passage outlets **558** at the trailing edges **512**.

[0186] The convergence angle or slope between the mouth and the throat, and the divergence angle or slope between the throat and the outlet may be specifically designed for maximizing efficiency during supersonic operation of the blade tips in which aerodynamic shock is generated as the airflow **504** is reduced in speed in the converging portion to choked flow of Mach 1 at the throat **556** followed in turn by subsonic diffusion in the diverging portion of the flow passage **552** from or aft of the throat **556** to the outlet **558**.

[0187] The ratio of the flow area at the passage outlet **558** over the flow area at the throat **556** is a conventional measure of effective camber of the airfoils **500**. The actual amount of airfoil camber near the tips **526** thereof may be increased slightly over a conventional turbofan design as indicated above to allow the turbofan to tolerate the lower tip solidity during part-speed operation.

[0188] As indicated above, a modern turbofan is designed for an operating range from takeoff to cruise to landing, with cruise operation being predominant and for which maximum efficiency and operability is desired. However, part-speed performance must also be considered in good turbofan design and accommodated by the higher camber introduced near the blade tips for the low solidity turbofan design.

[0189] As discussed herein, part-speed operability may be improved by increasing the camber of the airfoils **500** at the tips **526** thereof in conjunction with the reduction in solidity by reduction in blade count.

[0190] As improved efficiency of the fan may be obtained through lowering solidity, the turbofan design may itself be otherwise conventional except as modified in accordance with the present disclosure. For example, the airfoils **500** described herein may be relatively large in diameter for supersonic tip operation in a modern turbofan engine with a substantial pressure ratio of about 1.5. The corresponding bypass ratio of the fan air which bypasses the core engine may be about 7.5 or greater. Further, the airfoils may be provided with suitable aerodynamic sweep which is preferably forward or negative (S^-) at the tips **526** of the airfoils, with non-forward sweep near the hub or root.

[0191] To the extent one or more structures provided herein can be known in the art, it should be appreciated that the present disclosure can include combinations of structures not previously known to combine, at least for reasons based in part on conflicting benefits versus losses, desired modes of operation, or other forms of teaching away in the art.

[0192] This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

[0193] Further aspects are provided by the subject matter of the following clauses:

[0194] A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: $((FLL/FCL))/((SLL/SCL))$ to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

[0195] A turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor and a stator, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing

edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: $((FLL/FCL))/((SLL/SCL))$ to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$); wherein at least one of the first airfoil and the second airfoil is an airfoil that extends outwardly from an airfoil root and that includes opposite pressure and suction sides that extend longitudinally in span from the airfoil root to an airfoil tip, and that extend axially in chord between opposite airfoil leading and trailing edges, the airfoil further including stagger that increases between the airfoil root and the airfoil tip, camber that decreases therebetween, and chord length that increases outboard from the airfoil root to barrel the airfoil along both the airfoil leading and trailing edges, the airfoil further including forward aerodynamic sweep at the airfoil tip, and non-forward aerodynamic sweep between a maximum barrel in the airfoil and the airfoil root.

[0196] The turbine engine of any preceding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).

[0197] The turbine engine of any preceding clause, wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.

[0198] The turbine engine of any preceding clause, wherein the first stage of composite airfoils are fan blades.

[0199] The turbine engine of any preceding clause, wherein the second stage of composite airfoils are outlet guide vanes.

[0200] The turbine engine of any preceding clause, wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and wherein the second airfoil extends spanwise between a second root and a second tip to define a second span length.

[0201] The turbine engine of any proceeding clause, wherein the SPF is determined between 20% and 80% of the first span length and the second span length, inclusive of endpoints.

[0202] The turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.

[0203] The turbine engine of any proceeding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.

[0204] The turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq \text{SPF} \leq 2.5$).

[0205] The turbine engine of any proceeding clause, wherein the first sheath and the second sheath each have a first wall, a second wall, and a third wall interconnecting the first wall and the second wall.

[0206] The turbine engine of any proceeding clause, wherein the first wall, second wall, and third wall of the leading edge protector are oriented and shaped such that they define a U-shaped or C-shaped channel therebetween.

[0207] The turbine engine of any proceeding clause, wherein the channel is sized and shaped to receive the composite leading edge of the composite portion.

[0208] The turbine engine of any proceeding clause, wherein any of the first leading edge protector or the second leading edge protector are coupled to their corresponding composite portion at the corresponding composite leading edge to define at least one seam.

[0209] The turbine engine of any proceeding clause wherein the at least one seam is two seams on either side of the airfoil, and the corresponding first leading length or second leading length is measured from the corresponding leading edge to the seam furthest from the leading edge.

[0210] The turbine engine of any proceeding clause wherein the first leading length and the second leading length are measured from their corresponding leading edge to their corresponding seam.

[0211] The turbine engine of any proceeding clause wherein an amount of overlap between the first sheath or the second sheath and their corresponding.

[0212] A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by an expression: $((LL))/((CL))$ to define an airfoil protection factor (APF); and wherein the APF is greater than or equal to 0.1 and less than or equal to 0.3 ($0.1 \leq \text{APF} \leq 0.3$).

[0213] A turbine engine comprising: an engine core defining an engine centerline and comprising a rotor and a stator; a set of composite airfoils circumferentially arranged about

the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by an expression: $((LL))/((CL))$ to define an airfoil protection factor (APF); and wherein the APF is greater than or equal to 0.1 and less than or equal to 0.3 ($0.1 \leq \text{APF} \leq 0.3$); wherein at least one of the first airfoil and the second airfoil is an airfoil that extends outwardly from an airfoil root and that includes opposite pressure and suction sides that extend longitudinally in span from the airfoil root to an airfoil tip, and that extend axially in chord between opposite airfoil leading and trailing edges, the airfoil further including stagger that increases between the airfoil root and the airfoil tip, camber that decreases therebetween, and chord length that increases outboard from the airfoil root to barrel the airfoil along both the airfoil leading and trailing edges, the airfoil further including forward aerodynamic sweep at the airfoil tip, and non-forward aerodynamic sweep between a maximum barrel in the airfoil and the airfoil root.

[0214] The turbine engine of any proceeding clause, wherein the set of composite airfoils includes a first stage of composite airfoils and a second stage of composite airfoils downstream from the first stage of composite airfoils.

[0215] The turbine engine of any proceeding clause, wherein the first stage of composite airfoils is a set of fan blades and the second stage of composite airfoils is a set of outlet guide vanes.

[0216] The turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first airfoil protection factor (APF1) and the second stage of composite airfoils has a second airfoil protection factor (APF2).

[0217] The turbine engine of any proceeding clause, wherein the first airfoil protection factor (APF1) relates to the second airfoil protection factor (APF2) by an expression: $\text{APF1}/\text{APF2}$ to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq \text{SPF} \leq 4$).

[0218] The turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq \text{SPF} \leq 2.5$).

[0219] The turbine engine of any proceeding clause, wherein the set of composite airfoils extend spanwise between a root and a tip to define a span length and the APF is determined between 20% and 80% of the first span length and the second span length.

[0220] The turbine engine of any proceeding clause, wherein the leading edge protector overlaps with the composite leading edge to define a sheath.

[0221] The turbine engine of any proceeding clause, wherein the composite portion is formed from a polymer matrix composite (PMC).

[0222] The turbine engine of any proceeding clause, wherein the leading edge protector is a metallic leading edge protector.

[0223] An indirect drive turbine engine, comprising: an engine core defining an engine centerline and comprising a rotor defined by a fan including a plurality of fan blades rotatable about the engine centerline, a stator, a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, a speed reduction device driven by the turbine section for rotating the fan about the engine centerline, a first stage of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, a first airfoil of the first stage of composite airfoils comprising: a first composite portion extending chordwise between a first composite leading edge and a first trailing edge, a first leading edge protector comprising a first sheath receiving the first composite leading edge of the first composite portion, the first leading edge protector extending chordwise from a first leading edge towards the first composite portion for a first leading length (FLL), and the first composite portion and the first leading edge protector together defining an exterior surface of the first airfoil and extending chordwise between the first leading edge and the first trailing edge to define a first chord length (FCL), a second stage of composite airfoils located downstream of the first stage of composite airfoils and circumferentially arranged about the engine centerline, a second airfoil of the second stage of composite airfoils comprising: a second composite portion extending chordwise between a second composite leading edge and a second trailing edge, a second leading edge protector comprising a second sheath receiving the second composite leading edge of the second composite portion, the second leading edge protector extending chordwise from a second leading edge towards the second composite portion for a second leading length (SLL), and the second composite portion and the second leading edge protector together defining an exterior surface of the second airfoil and extending chordwise between the second leading edge and the second trailing edge to define a second chord length (SCL), wherein the first leading length (FLL) and the first chord length (FCL) relate to the second leading length (SLL) and the second chord length (SCL) by an expression: $((FLL/FCL)/((SLL/SCL)))$ to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

[0224] The indirect drive turbine engine of any preceding clause, wherein the speed reduction device is a power gearbox.

[0225] The indirect drive turbine engine of any preceding clause, wherein the turbine section includes a fan drive turbine and a second turbine and the second turbine is disposed forward of the fan drive turbine and the fan drive turbine includes a plurality of fan drive turbine stages with the speed reduction device driven by the fan drive turbine.

[0226] The indirect drive turbine engine of any preceding clause, further comprising a fan casing or nacelle.

[0227] The indirect drive turbine engine of any preceding clause, wherein the fan drive turbine has between 3 and 5 stages.

[0228] The indirect drive turbine engine of any preceding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 2:1 and 5:1.

[0229] The indirect drive turbine engine of any preceding clause, wherein a bypass ratio is between 10:1 and 22:1.

[0230] The indirect drive turbine engine of any preceding clause, wherein a fan blade tip speed of the fan is less than 1400 feet per second.

[0231] The indirect drive turbine engine of any preceding clause, wherein the core is an open rotor engine.

[0232] The indirect drive turbine engine of any preceding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 6:1 and 12:1.

[0233] The indirect drive turbine engine of any preceding clause, wherein a bypass ratio is between 25:1 and 125:1.

[0234] The indirect drive turbine engine of any preceding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils include a polymer matrix composite (PMC).

[0235] The indirect drive turbine engine of any preceding clause, wherein at least one of the first leading edge protector or the second leading edge protector is a metallic leading edge protector.

[0236] The indirect drive turbine engine of any preceding clause, wherein the first stage of composite airfoils include the plurality of fan blades.

[0237] The indirect drive turbine engine of any preceding clause, wherein the second stage of composite airfoils are outlet guide vanes or fan guide vanes.

[0238] The indirect drive turbine engine of any preceding clause, wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and wherein the second airfoil extends spanwise between a second root and a second tip to define a second span length.

[0239] The indirect drive turbine engine of any preceding clause, wherein the SPF is determined between 20% and 80% of the first span length and the second span length, inclusive of endpoints.

[0240] The indirect drive turbine engine of any preceding clause, wherein the first stage of composite airfoils has a first number of airfoils and the second stage of composite airfoils has a second number of airfoils and the first number is different than the second number.

[0241] The indirect drive turbine engine of any preceding clause, wherein the first stage of composite airfoils and the second stage of composite airfoils are configured to rotate.

[0242] The indirect drive turbine engine of any preceding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).

[0243] The indirect drive turbine engine of any preceding clause, wherein the first sheath and the second sheath each have a first wall, a second wall, and a third wall interconnecting the first wall and the second wall.

[0244] The indirect drive turbine engine of any preceding clause, wherein the first wall, second wall, and third wall of the leading edge protector are oriented and shaped such that they define a U-shaped or C-shaped channel therebetween.

[0245] The indirect drive turbine engine of any preceding clause, wherein the channel is sized and shaped to receive the composite leading edge of the composite portion.

[0246] The indirect drive turbine engine of any preceding clause, wherein any of the first leading edge protector or the second leading edge protector are coupled to their corresponding composite portion at the corresponding composite leading edge to define at least one seam.

[0247] The indirect drive turbine engine of any preceding clause wherein the at least one seam is two seams on either side of the airfoil, and the corresponding first leading length

or second leading length is measured from the corresponding leading edge to the seam furthest from the leading edge.

[0248] The indirect drive turbine engine of any proceeding clause wherein the first leading length and the second leading length are measured from their corresponding leading edge to their corresponding seam.

[0249] The indirect drive turbine engine of any proceeding clause wherein an amount of overlap between the first sheath or the second sheath and their corresponding.

[0250] An indirect drive turbine engine comprising: an engine core defining an engine centerline and comprising a rotor defined by a fan including a plurality of fan blades rotatable about the engine centerline, a stator, a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, a speed reduction device driven by the turbine section for rotating the fan about the engine centerline; a set of composite airfoils circumferentially arranged about the engine centerline and defining at least a portion of the rotor, an airfoil of the set of composite airfoils comprising: a composite portion extending chordwise between a composite leading edge and a trailing edge; a leading edge protector coupled to the composite portion at the composite leading edge to define a seam, and extending chordwise between a leading edge and the seam to define a leading length (LL); and the composite portion and the leading edge protector together defining an exterior surface of the airfoil and extending chordwise between the leading edge and the trailing edge to define a chord length (CL); wherein the leading length (LL) and the chord length (CL) relate to each other by an expression: $((LL))/((CL))$ to define an airfoil protection factor (APF); and wherein the APF is greater than or equal to 0.1 and less than or equal to 0.3 ($0.1 \leq APF \leq 0.3$).

[0251] The indirect drive turbine engine of any proceeding clause, wherein the turbine section includes a fan drive turbine and a second turbine and the second turbine is disposed forward of the fan drive turbine and the fan drive turbine includes a plurality of fan drive turbine stages with the speed reduction device driven by the fan drive turbine.

[0252] The indirect drive turbine engine of any proceeding clause, further comprising a fan casing or nacelle.

[0253] The indirect drive turbine engine of any proceeding clause, wherein the fan drive turbine has between 3 and 5 stages.

[0254] The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 2:1 and 5:1.

[0255] The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 10:1 and 22:1.

[0256] The indirect drive turbine engine of any proceeding clause, wherein a fan blade tip speed of the fan is less than 1400 feet per second.

[0257] The indirect drive turbine engine of any proceeding clause, wherein the core is an open rotor engine.

[0258] The indirect drive turbine engine of any proceeding clause, wherein the speed reduction device is a power gearbox having a power gearbox reduction ratio between 6:1 and 12:1.

[0259] The indirect drive turbine engine of any proceeding clause, wherein a bypass ratio is between 25:1 and 125:1.

[0260] The indirect drive turbine engine of any proceeding clause, wherein the set of composite airfoils includes a first

stage of composite airfoils and a second stage of composite airfoils downstream from the first stage of composite airfoils.

[0261] The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils is the plurality of fan blades and the second stage of composite airfoils is a set of outlet guide vanes or a set of fan guide vanes.

[0262] The indirect drive turbine engine of any proceeding clause, wherein the first stage of composite airfoils has a first airfoil protection factor (APF1) and the second stage of composite airfoils has a second airfoil protection factor (APF2).

[0263] The indirect drive turbine engine of any proceeding clause, wherein the first airfoil protection factor (APF1) relates to the second airfoil protection factor (APF2) by an expression: $APF1/APF2$ to define a stage protection factor (SPF), wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

[0264] The indirect drive turbine engine of any proceeding clause, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq SPF \leq 2.5$).

[0265] The indirect drive turbine engine of any proceeding clause, wherein the set of composite airfoils extend spanwise between a root and a tip to define a span length and the APF is determined between 20% and 80% of the first span length and the second span length.

[0266] The indirect drive turbine engine of any proceeding clause, wherein the leading edge protector overlaps with the composite leading edge to define a sheath.

[0267] The indirect drive turbine engine of any proceeding clause, wherein the composite portion is formed from a polymer matrix composite (PMC).

[0268] The indirect drive turbine engine of any proceeding clause, wherein the leading edge protector is a metallic leading edge protector.

[0269] The turbine engine of any proceeding clause, wherein the turbine engine is an indirect turbine engine comprising a compressor section, a combustor in fluid communication with the compressor section, a turbine section in fluid communication with the combustor, and a speed reduction device driven by the turbine section, wherein the first airfoil extends spanwise between a first root and a first tip to define a first span length and the second airfoil extends spanwise between a second root and a second tip to define a second span length less than the first span length.

[0270] The turbine engine of any proceeding clause, wherein the forward aerodynamic sweep is a maximum forward aerodynamic sweep that transitions to maximum aft sweep inwardly along the airfoil leading edge, and that decreases in aft magnitude to the non-forward aerodynamic sweep at the airfoil root.

[0271] The turbine engine of any proceeding clause, wherein the airfoil leading edge is a sharp airfoil leading edge and the airfoil trailing edge is a sharp airfoil trailing edge from the airfoil root to the airfoil tip, and wherein the airfoil gradually increases to a maximum thickness along a midchord region therebetween.

[0272] The turbine engine of any proceeding clause, wherein the airfoil further includes low sweep near the airfoil root that is less than about half a maximum forward aerodynamic sweep in the airfoil thereabove.

[0273] The turbine engine of any proceeding clause, wherein the camber has a greater rate of decrease outboard

from the root into the barrel than outboard from the barrel toward the tip for locally increasing the camber near the tip

[0274] The turbine engine of any proceeding clause, wherein the maximum forward sweep at the airfoil tip transitions to aft sweep inwardly therefrom in the airfoil barrel.

[0275] The turbine engine of any proceeding clause, wherein the airfoil includes zero sweep along the leading edge at the root.

[0276] The turbine engine of any proceeding clause, wherein the airfoil includes a local peak in aft sweep along the leading edge between the root and the maximum aft sweep.

[0277] The turbine engine of any proceeding clause, wherein the airfoil camber is defined by the difference in axial inlet angle at the leading edge and axial exit angle at the trailing edge, and the exit angle increases from root to tip to correspondingly decrease the camber from the root toward the tip.

[0278] A plurality of fan blades of any proceeding clause, wherein a plurality of the fan blades are arranged circumferentially in a row to define corresponding flow passages between adjacent airfoils for pressurizing air; each of the airfoils including stagger increasing between the root and the tip to position the leading edge of one airfoil circumferentially adjacent to the suction side of the next adjacent airfoil to define a mouth for the flow passage therebetween, with the flow passage converging to a throat aft from the mouth; adjacent airfoils including a sloped platform conforming to the sloped root for effecting with the airfoils the non-forward sweep near the platform; and wherein the camber has a local increase between the tip and barrel.

[0279] A fan blade row of any proceeding clause, wherein the airfoil stagger positions the trailing edge of one airfoil circumferentially adjacent to the pressure side of the next adjacent airfoil to define an outlet for the flow passage therebetween; and the airfoil tips vary in thickness between the leading and trailing edges to diverge the flow passage therebetween.

[0280] The turbine engine of any proceeding clause, wherein the camber varies in slope with a local negative peak immediately below the tip followed by a local positive peak therebelow.

[0281] The turbine engine of any proceeding clause, wherein the camber has a local increase between the tip and barrel.

[0282] A fan blade row of any proceeding clause comprising no more than twenty of the fan blades having a solidity defined by the ratio of the airfoil chord over the circumferential pitch and being low in magnitude at the tips to position the leading edge of each tip circumferentially near the trailing edge of the next adjacent tip and correspondingly increase the width of the throat.

[0283] A method of improving aerodynamic efficiency of a fan blade according to any proceeding clause comprising: deriving the blade from a preexisting fan blade; increasing the airfoil camber locally near the airfoil tip; and eliminating forward sweep along the leading edge near the airfoil root.

What is claimed is:

1. A turbine engine comprising:

a fan;

a turbomachine comprising a compressor section, a combustor in fluid communication with the compressor section, and a turbine section in fluid communication with the combustor;

a speed reduction device driven by the turbine section and configured for rotating the fan; and

composite airfoils comprising at least one composite fan blade and at least one composite outlet guide vane downstream from the at least one composite fan blade;

wherein an airfoil of the composite airfoils comprises:

a composite body extending chordwise from a body leading edge to a body trailing edge, and

a leading edge protector having a protector leading edge different from the body leading edge, the leading edge protector receiving at least a portion of the body leading edge, and wherein the leading edge protector comprises a first wall defining a first length and defining the end of the leading edge protector, a second wall defining a second length shorter than the first length, and a third wall interconnecting the first wall and the second wall and defining the protector leading edge;

wherein a leading length (LL) extends chordwise from the protector leading edge to an end of the leading edge protector, wherein the first wall or the second wall defines the end, and a chord length (CL) extends chordwise from the protector leading edge to the body trailing edge, and

wherein the LL is related to the CL by an airfoil protection factor

$$(APF) = \frac{(LL)}{(CL)};$$

and a first APF (APF1) for the at least one composite fan blade is greater than or equal to 0.2 and less than or equal to 0.30 ($0.2 \leq APF1 \leq 0.3$) and a second APF (APF2) for the at least one composite outlet guide vane is greater than or equal to 0.08 and less than or equal to 0.17 ($0.08 \leq APF2 \leq 0.17$).

2. The turbine engine of claim 1, wherein the first wall is provided on a pressure side of the airfoil.

3. The turbine engine of claim 1, wherein the LL ranges from 1.5 cm to 19 cm.

4. The turbine engine of claim 3, wherein the CL ranges from 9.3 cm to 77 cm.

5. The turbine engine of claim 4, wherein the airfoil is the at least one composite fan blade.

6. The turbine engine of claim 5, wherein the composite body includes a polymer matrix composite and the leading edge protector is a metallic leading edge protector.

7. The turbine engine of claim 1, wherein the APF1 relates to the APF2 by an expression:

$$\frac{APF1}{APF2}$$

to define a stage protection factor (SPF), and wherein the SPF is greater than or equal to 0.7 and less than or equal to 4 ($0.7 \leq SPF \leq 4$).

8. The turbine engine of claim 7, wherein the SPF is greater than or equal to 0.95 and less than or equal to 2.5 ($0.95 \leq \text{SPF} \leq 2.5$).

9. The turbine engine of claim 7, wherein the at least one composite fan blade extends spanwise between a first root and a first tip to define a first span length and the at least one composite outlet guide vane extends spanwise between a second root and a second tip to define a second span length and the SPF is determined at a location between 20% and 80% of the first span length and the second span length, inclusive of endpoints.

10. The turbine engine of claim 1, wherein the first wall, the second wall, and the third wall of the leading edge protector define a U-shaped or C-shaped channel, and wherein the channel receives the composite leading edge.

11. The turbine engine of claim 1, wherein the speed reduction device is a power gearbox.

12. The turbine engine of claim 1, wherein a bypass ratio is from 10:1 to 22:1.

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